

Blockchain Integration in VANETs: Challenges and Research Frontiers in Security and Trust

Abstract

Vehicular Ad-hoc Network (VANET), an emerging and powerful technology, has become a trending research topic in the realm of Intelligent Transportation Systems (ITS) that has the potential to revolutionize the transportation system in numerous ways. Modern-day vehicles in VANET are equipped with sophisticated onboard processing units and assistant features that improve traffic safety and efficiency. These capabilities enable the dissemination of safety messages among vehicles across the network, which was otherwise impossible using traditional means. However, the dynamic network topology and the highly mobile nature of VANET bring forth their own set of challenges related to privacy, authentication, trust, and security issues among the participating vehicles. In order to overcome these challenges, some of the existing works incorporated blockchain technology due to its essential characteristics, such as decentralization, transparency, tamper-proof nature, and auditing facilities that aim to make VANET more robust against the challenges mentioned above. In this comprehensive study, we accumulated and classified the existing research works done in the field of authentication, trust, security, and privacy in blockchain-based VANET and presented them in a well-defined, categorized collection with extensive analysis of the works by means of tables and insightful visualizations. The main focus of the study is to build a profound understanding and give the readers a broader perspective on the research area. Additionally, the study provides insightful direction on open issues that may implicate potential research endeavors for researchers on the topic.

Keywords: VANET, blockchain, privacy, security, trust, authentication

1. Introduction

The rapid increase in the number of vehicles has made traffic congestion as well as road accidents a major concerning issue for transportation experts around the world [1]. Hence, the concept of an Intelligent Transportation System (ITS) has gained significant traction as a means to build smart cities and reduce transportation anomalies. One of the most advanced technologies that facilitates the development of ITS is the Vehicular Ad-hoc Network (VANET). As illustrated in figure 1, VANET is a collaborative network of rapidly moving, self-organizing vehicles that use Dedicated Short-Range Communication (DSRC) for Vehicle-to-Vehicle (V2V) or Vehicle-to-Infrastructure (V2I) communication. Critical safety messages related to traffic and road conditions are shared among vehicles or between vehicles and Roadside Units (RSUs) that provide smart driving assistance, reduce road incidents, and improve traffic congestion issues, leading to a better and more comfortable driving experience. It can further help Law Enforcement Agencies (LEA) and transportation authorities to reconstruct the events of accidents for rapid investigation [2]. Road-side units (RSUs) are deployed as supporting infrastructures equipped with computational units and wireless communication modules to facilitate long-range communication and extension of the network. The vehicles are fitted with Onboard Units (OBUs) having powerful processors, cameras, sensors, and wireless communication modules (though typically battery-constrained, which limits sustained computation-intensive operations), which enable the car manufacturers to build vehicles with sophisticated features like lane tracking, alert systems, and intelligent infotainment [3].

Although the use of VANET has unlocked several striking features contributing to the advancement of ITS, its dynamic nature and the rapidly moving network of vehicles bring forward a multifaceted set of challenges that must be addressed for successful real-world deployment. Scalability remains a fundamental concern, as VANETs must support communication among potentially thousands of vehicles within dense urban environments while maintaining acceptable performance levels [4]. Reliability concerns emerge from the highly mobile topology and intermittent connectivity patterns, where vehicles frequently enter and exit communication range, making consistent message delivery difficult to guarantee [5]. Beyond these infrastructure-level concerns, VANETs face critical challenges related to privacy, trust management, and security among participating vehicles, a constellation of intercon-

nected impediments that form the core focus of this survey.

The privacy, trust, and security challenges are particularly critical because they directly impact the safety and integrity of vehicular communication. Vehicles in a VANET are utterly unknown to each other, creating an environment where a malicious vehicle may enter the network and disseminate forged messages to gain its own benefit at the cost of endangering traffic safety and even causing fatal road accidents. In a less overt scenario, they may operate passively, intercepting and analyzing network messages to trace a vehicle's route or steal private information in order to orchestrate future security and privacy attacks. Therefore, it is essential to ensure the authenticity of vehicles (authentication), assess the reliability of disseminated information (trust management), protect against malicious attacks (security), and preserve the identity and location confidentiality of legitimate users (privacy). These four concerns are inherently intertwined: authentication mechanisms must preserve privacy, trust evaluation depends on secure message verification, and privacy-preserving schemes must not undermine security guarantees. While scalability and reliability affect the performance and operational efficiency of VANETs, compromised trust, security, and privacy can have immediate life-threatening consequences, making them paramount concerns [2, 4, 6]. Effective countermeasures to address these authentication, trust, security, and privacy challenges are necessary to ensure the widespread deployment of VANETs.

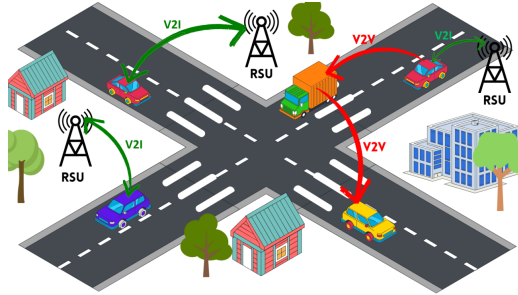


Figure 1: General Architecture of VANET

Various solutions have been proposed by existing works to resolve the issues of authentication, trust management, privacy, and security in VANETs. Traditionally, securing VANETs has relied on cryptographic mechanisms, with digital signatures serving as the foundational tool providing essential authentication (proof of origin) and data integrity (proof of no tampering).

The primary architectural challenge lies in managing the cryptographic keys for these signatures, leading to different frameworks such as Public Key Infrastructure (PKI), Identity-Based Signatures (IBS), and Certificateless Signatures (CLS) [7, 8]. Alongside authentication, reputation-based trust management systems have been employed to assess the reliability of message sources through scoring mechanisms [9]. However, these approaches suffered from critical and overlapping deficiencies. Their reliance on centralized Trusted Third Parties (TTPs) creates both a single point of failure (SPoF) and severe scalability bottlenecks, particularly in managing certificate and pseudonym revocation required for privacy, as well as reputation score storage and updates for trust evaluation [10, 11, 12]. Furthermore, they impose high computational overheads, as the frequent signature verification in dense traffic conflicts with the low-latency demands of safety applications [13]. The centralized systems also lack the inherent transparency and auditable non-repudiation needed to enforce accountability and build trust among mutually mistrusting entities, making them vulnerable to manipulation of reputation scores by compromised authorities [12].

To overcome these centralized limitations and foster trust among mutually mistrusting entities, blockchain-based solutions have particularly gained attention due to their valuable intrinsic properties [14]. Blockchain provides decentralization by allowing records to be held redundantly by every participant, thus eliminating the SPoF [11]. Its use of cryptographic chaining ensures immutability and a tamper-proof nature for shared data, upholding integrity and non-repudiation. Moreover, the technology enhances transparency and provides essential auditing facilities through its shared ledger and consensus mechanisms [14, 15]. This potential, however, is contingent on adaptation, as conventional blockchain architectures are insufficient for VANETs. Their high-latency consensus mechanisms and inherent transparency directly conflict with the real-time performance (low latency, high throughput) and strict confidentiality requirements of vehicular applications. Therefore, effective deployment requires integrating blockchain's core strengths with VANET-specific adaptations, such as lightweight consensus protocols (e.g., proof-of-authority) and robust privacy-preserving mechanisms (e.g., consortium models), to meet these dynamic environmental constraints [12, 14]. Our review paper aims to provide an overview of the existing studies that have utilized this integrated blockchain technology to address the deficiencies pertaining to authentication, trust, security, and privacy in VANET.

1.1. Scope of This Study

This survey is centered on the intersection of blockchain technology and VANETs, with a particular emphasis on how blockchain is leveraged to address foundational concerns in security, privacy, authentication, and trust management. Rather than presenting a general overview of blockchain in intelligent transportation systems, we focus exclusively on VANET-specific architectures and challenges. To ensure relevance and depth of our survey, we have applied a well-defined inclusion strategy.

This study includes

- Peer-reviewed studies that incorporate blockchain into V2V and V2I communications within VANETs.
- Research proposing original system architectures, protocols, or mechanisms that utilize core blockchain properties such as decentralization, immutability, and transparency through distributed ledgers or consensus mechanisms, not merely as generic storage solutions.
- Papers that provide empirical, simulation-based, or theoretical evaluations of blockchain’s applicability to VANET environments.
- Works on broader vehicular ecosystems (Internet of Vehicles, Cooperative ITS) only when explicitly designed for and evaluated in VANET-specific settings with V2V/V2I communication patterns.

By defining this scope, the study aims to serve as a targeted reference for researchers exploring blockchain as a viable solution for the dynamic and constrained environment of VANETs.

Differentiation from Existing Surveys. Previous surveys on blockchain in vehicular networks exhibit several critical limitations, detailed in Tables 1 and 2. Traditional VANET security surveys focus on centralized trust models without exploring blockchain’s decentralized capabilities, while blockchain-enabled surveys either adopt overly broad scope (covering general ITS or C-ITS rather than VANET-specific constraints) or classify works by application area and network type rather than functional mechanism. These approaches obscure how specific blockchain properties address particular VANET challenges and provide limited systematic evaluation of mobility, latency, and overhead requirements essential for practical deployment.

This survey distinguishes itself through three key contributions. First, we adopt a functionality-based taxonomy classifying solutions by technical

purpose (handover authentication, batch verification, pseudonymity management, attack prevention) rather than application domains. Second, we employ a dual-table structure pairing conceptual overview with implementation feasibility assessment, explicitly evaluating each solution against VANET-specific constraints. Third, we provide critical technical analysis through dedicated subsections identifying common pitfalls, synthesize cross-cutting patterns in Section 6, and operationalize findings into testable research questions in Section 7. This structure complements existing reviews with deeper insight into blockchain’s practical viability in vehicular environments.

1.2. Our Contribution

This paper presents a focused and in-depth survey of blockchain-based approaches (from 55 academic papers) to address four critical concerns in VANETs: authentication, trust management, security, and privacy. Unlike prior surveys that offer broad overviews or lack domain-specific structure, our work aims to provide newcomers and researchers in the field with a comprehensive, structured, and insightful analysis. Our key contributions are as follows:

- **Domain-Specific and Functionality-Based Taxonomy.** We introduce a well-organized taxonomy that classifies blockchain-enabled VANET solutions not by broad application domains or network types, but by functionality, including handover authentication, group-based trust evaluation, batch verification, pseudonymity, secure message transmission, and attack prevention. This approach offers a clearer understanding of how blockchain specifically addresses core VANET challenges.
- **Systematic Evaluation Against VANET Constraints.** We critically analyze how existing blockchain solutions meet the unique demands of VANET environments, particularly in terms of mobility, latency, and communication overhead. These essential performance constraints are often underexplored in previous literature, and our study brings them to the forefront to assess the practicality of blockchain adoption in real-world vehicular scenarios.
- **Comprehensive Mapping for Benchmarking and Reproducibility.** We provide detailed tabular mappings of tools, consensus mechanisms, blockchain architectures, and performance metrics used in var-

ious studies. This structured comparison aids reproducibility, benchmarking and offers a quick reference for researchers to evaluate or replicate existing approaches.

- **Cross-Cutting Analysis and Overlapping Categorizations.** We recognize that several solutions span multiple subdomains, for instance, schemes that integrate both trust evaluation and authentication. Accordingly, we have included overlapping categorizations with appropriate cross-referencing, enabling readers to understand interdependencies and multi-faceted solutions.
- **Future Directions and Research Opportunities.** Building upon the challenges identified in the literature, we propose contextualized research directions that reflect VANET-specific constraints and emerging trends. These insights aim to guide future work by identifying unresolved issues and highlighting promising areas for innovation in blockchain-based vehicular security.

1.3. Organization of the Paper

The rest of the paper is organized as follows: Section 2 discusses our approach for collecting and filtering different papers for this review, thus serving as a general guideline to readers on how to start exploring a new research topic. Section 3 presents the existing survey papers related to authentication, trust, privacy, and security in VANETs. Afterward, a brief background study on blockchain-based VANET is provided in section 4, primarily containing the foundation for new readers to develop the mindset to analyze and evaluate blockchain-based VANET schemes. Consequently, section 5 presents the categorized review of the collected papers through a systematic breakdown into further subsections. Readers are encouraged to go through the limitations listed at the end of every subgroup of the papers in this section. Subsequently, section 6 notes the main challenges with a similar flow as seen in section 5, but, in a supplementary manner, accompanied by the open discussion on the existing practices and trends. Based on the discussion, section 7 provides directional remarks for researchers for potential contributions and future research endeavors in the domain of blockchain-based solutions for authentication, trust, privacy, and security in VANETs. Finally, section 8 concludes this work. A graphical outline of the paper is presented through figure 2.

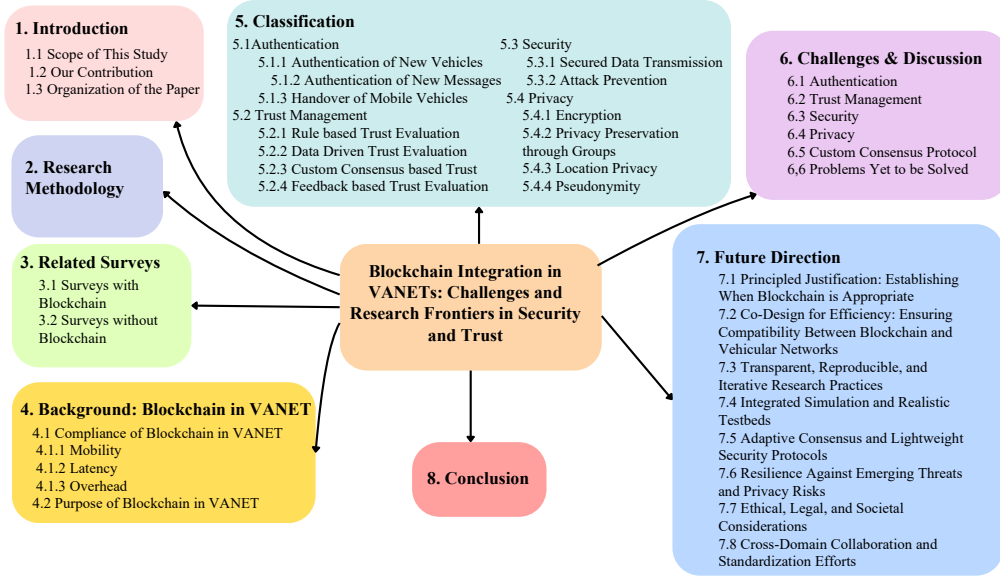


Figure 2: Overview of the Survey Paper

Frequently Used Acronyms

The frequently used acronyms of this paper are listed in alphabetical order for the reader's convenience.

CA: Certification Authority
 CRL: Certificate Revocation List
 DSRC: Dedicated Short Range Communication
 ECC: Elliptic Curve Cryptography
 EM: Emergency Message
 IoV: Internet of Vehicles
 IPP: Identity Privacy Preservation
 ITS: Intelligent Transportation System
 IVC: Inter-Vehicle Communication (similar to V2V)
 KDC: Key Distribution Center (similar to TA)
 LBS: Location-Based Service
 LEA: Law Enforcement Agency
 LPP: Location Privacy Preservation
 OBU: On Board Unit
 PKC: Public Key Cryptography

PKI: Public Key Infrastructure
RSU: Road Side Unit
SBM: Security Beacon Message
SCK: Symmetric Key Cryptography
SDN: Software Defined Network
TA: Trusted Authority
TPD: Tamper-proof Device
TTP: Trusted Third Party
V2I: Vehicle to Infrastructure
V2V: Vehicle to Vehicle

2. Research Methodology

Our study conducted a Systematic Literature Review (SLR) following a conventional and methodical approach [16]. A combination of the listed keywords was applied to search for the relevant articles that mainly focused on blockchain-based works on authentication, trust management, privacy, and security in VANETs. Searches were conducted in December 2024 using both Google Scholar as a search engine and the native search features of individual academic repositories, including IEEE Xplore, Scopus, SpringerLink, ACM Digital Library, MDPI, ScienceDirect, and Wiley to ensure comprehensive coverage (illustrated in figure 3). The initial systematic search covering publications from 2018 to the end of 2024 yielded 3,241 results. After the removal of 1,363 duplicates, a total of 1,878 articles were obtained.

Stage 1: Title and Abstract Screening. An initial screening was carried out where the title and abstract of the 1,878 retrieved articles were examined to identify studies with potential relevance to blockchain applications in VANETs. Based solely on title and abstract content, 1,708 articles were excluded for lacking clear relevance to the study scope, resulting in 170 potentially relevant articles for further evaluation.

Stage 2: Dual-Author Quality Assessment. To enforce the scope of our study and ensure methodological rigor and quality assessment, each of the 170 articles was independently screened by both the first and second authors. At this stage, reviewers examined the abstract, full introduction section, stated contributions, and research methodology section of each article. Articles were excluded if they demonstrated insufficient methodological rigor, lacked substantive contributions to the field, had inadequate evalua-

tion or implementation details, or did not align with blockchain applications in VANETs. This dual-author screening process ensured consistent quality assessment while avoiding bias from citation-based metrics. A total of 87 articles were retained after this screening phase, while 83 articles were excluded.

Stage 3: Full-Text Review for Final Eligibility. The 87 retained articles then underwent a comprehensive full-text review by the authors to verify final alignment with the inclusion criteria stated in Section 1.1. This stage involved thorough examination of complete article contents to assess aspects such as the depth and novelty of blockchain integration in VANET contexts, comprehensiveness of security or privacy mechanisms, quality of experimental validation, and overall study quality—details not fully discernible from abstracts and methodology sections alone. This rigorous full-text evaluation resulted in the exclusion of 32 additional articles, yielding a final selection of 55 articles.

The pie chart in Figure 3 illustrates the distribution of the final 55 surveyed articles by their source repository. Consequently, repositories such as ACM and Wiley are not included in the chart, as all articles retrieved from them were identified as duplicates of those in other databases. Furthermore, Google Scholar is omitted, as it was utilized as a search engine for discovery rather than as a distinct repository in the final distribution analysis.

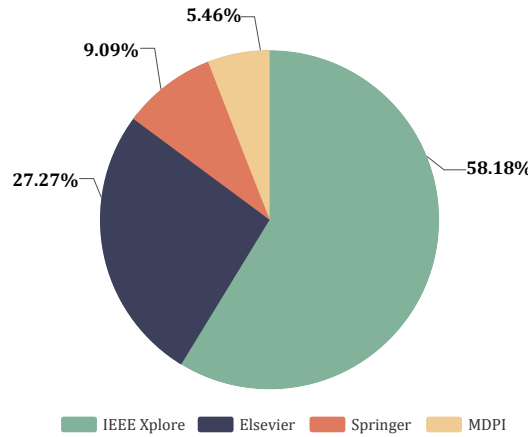


Figure 3: Distribution of articles with respect to repositories

3. Related Surveys

The inclusion of survey studies is of utmost importance in our pursuit of a full understanding of authentication, trust, security, and privacy in VANET. By including both conventional VANET and emerging blockchain-based VANET paradigms, we are able to provide a comprehensive perspective of the progressing research domain. This meticulously curated selection of survey and review papers plays a crucial role in providing guidance and direction in navigating the complex difficulties and potential solutions within well-established and emerging vehicular networking systems. Our objective is to extract invaluable insights that emphasize the crucial significance of protecting communication and data integrity within this ever-changing field. Our analyses of these works are presented in Sections 3.1 and 3.2, with corresponding summaries in Tables 1 and 2.

3.1. *Surveys without Blockchain*

Traditional VANET security surveys established foundational understanding without considering blockchain’s potential. [8] surveyed security attacks, characteristics, and countermeasures, proposing a five-level global security architecture. [17] examined pseudonym mechanisms based on public key cryptography, group signatures, and symmetric authentication, proposing an abstract pseudonym lifecycle and categorization. More recently, [22] provided fine-grained analysis of pseudonym change strategies (PCSs), classifying 52 schemes across seven categories with comparative evaluation of operational mechanisms, attacker models, and privacy metrics. [19] outlined security and privacy requirements for location-based services (LBSs), reviewing evaluation metrics and enhancement approaches. [21] categorized solutions by cryptographic mechanisms (PKI, IBC), comparing over 50 security schemes across authentication, privacy, and implementation dimensions. [20] classified authentication and privacy schemes by mechanism: pseudonymous-based (PAPS), group signature-based (GAPS), ID-PKC-based (IAPS), and hybrid anonymous-based (HAPS), comparing strengths, limitations, and performance parameters. [9] surveyed location privacy, security, and trust management, notably reviewing network simulators and platforms for VANET research. [3] provided comprehensive categorization of privacy preservation techniques and authentication schemes. Table 1 summarizes these surveys and their research gaps.

Table 1: Survey papers on security, trust, authentication, and privacy in VANETs (without blockchain)

Paper	Year	Area of Survey	Research Gaps
[8]	2014	Prevalent attacks, analysis, traits, and remedies	Limited exploration of decentralized solutions; lacks systematic evaluation frameworks; missing integration of emerging trust management paradigms
[17]	2015	Pseudonym schemes for privacy preservation	No consideration of blockchain-based decentralized pseudonym management; lacks analysis of scalability in high-mobility scenarios; missing comparative performance evaluation
[18]	2017	Secured authentication schemes	Focuses on traditional centralized PKI approaches; does not address distributed ledger alternatives; limited discussion of handover authentication in mobile environments
[13]	2017	Security and privacy issue	Broad overview without depth on specific mechanisms; lacks blockchain integration for tamper-proof auditing; missing VANET-specific performance constraints analysis
[2]	2017	Security challenges and solutions in VANET	Primarily centralized trust models; no exploration of distributed consensus mechanisms; lacks discussion of immutability for non-repudiation
[19]	2018	Security and privacy for LBS in vehicular communications	Location privacy solutions rely on centralized servers; does not explore blockchain-based transparency for accountability; missing analysis of decentralized cloaking techniques
[20]	2019	Authentication and privacy schemes classification	Classification based on cryptographic primitives only; lacks functionality-based taxonomy; no consideration of blockchain-enabled distributed authentication
[3]	2020	Secure authentication and privacy-preserving techniques	Traditional approaches without blockchain integration; limited discussion of decentralized revocation mechanisms; missing evaluation of consensus-based trust
[21]	2020	Cryptographic mechanisms, trust management, and intrusion detection systems	Comprehensive but lacks blockchain perspective; centralized trust management models; no exploration of smart contract automation for security protocols
[22]	2025	Pseudonym change strategies for location privacy preservation	Fine-grained PCS classification but no blockchain-based pseudonym lifecycle management; lacks decentralized revocation strategies; missing distributed storage and synchronization analysis

Table 2: Survey papers related to security, trust, authentication, and privacy in VANETs (with blockchain)

Paper	Year	Area of Survey	Research Gaps
[12]	2021	Blockchain for security, privacy, and trust in vehicular network	Classification by network type (VANET, VDTN, IoV) rather than functional mechanisms; lacks depth in VANET-specific performance constraints (mobility, latency, overhead); no systematic evaluation framework; limited technical critique of implementation feasibility
[11]	2022	Blockchain-based VANETs	Six major categories with significant overlaps and uneven distribution; high-level observations without systematic technical analysis; lacks detailed implementation feasibility assessment; missing comprehensive benchmarking framework
[23]	2022	Blockchain for securing vehicular networks (IoV, EVs, applications)	Broad vehicular ecosystem focus without VANET-specific depth; organized by application domains and attack types rather than functional mechanisms; lacks emphasis on V2V/V2I communication constraints; missing VANET mobility and latency analysis
[14]	2022	Blockchain for VANET security (authentication, trust, attacks)	Mechanism-centric (key management, access control) but lacks functionality-based taxonomy; insufficient analysis of group-based vs. non-group authentication trade-offs; limited discussion of consensus protocol suitability for high-mobility scenarios
[24]	2022	Application of blockchain-based ITS	Extensive background on technologies but trivial literature classification; broad ITS scope dilutes VANET-specific insights; lacks critical evaluation of blockchain necessity vs. alternatives; missing performance validation analysis
[15]	2023	Blockchain-based Co-operative ITS application	Focuses on C-ITS rather than pure VANETs; qualitative comparison without quantitative performance metrics; lacks structured evaluation against VANET constraints (mobility, latency, overhead); missing reproducibility guidelines and standardization discussion

3.2. Surveys with Blockchain

Blockchain-enabled vehicular network surveys exhibit varied approaches with distinct limitations detailed in Table 2. [12] compared blockchain-based privacy, trust management, and security models but classified works by application area (VANET, VDTN, IoV, VSN) rather than functional approach, providing limited depth and lacking concrete guidance on trends and open issues. [11] identified six major categories with significant overlaps and uneven distribution, offering high-level observations without systematic technical analysis. [23] surveyed blockchain security schemes across the broader vehicular ecosystem (IoV, electric vehicle sharing, autonomous systems), categorizing solutions by security attacks and application domains. In contrast, [14] focused specifically on VANETs, structuring analysis around security mechanisms (attacks, key management, access control) for highly mobile V2V and V2I communication. The VANET authentication literature commonly distinguishes between group-based schemes (shared or group public keys for collective authentication) and non-group-based schemes (individual vehicle keys), fundamental to understanding privacy-performance trade-offs. [24] surveyed blockchain in ITS and blockchain-based IoVs but devoted substantial space to background knowledge with trivial literature classification. [15] focused on Cooperative Intelligent Transport Systems (C-ITS), highlighting adoption challenges (delay, mobility) and reviewing blockchain-based VANET works in four categories with qualitative comparison and insightful discussion of communication and security issues. Table 2 provides comprehensive comparison of these surveys and identifies their specific research gaps.

4. Background: Blockchain in VANET

Since the advent of blockchain technology, researchers and practitioners across several domains have been using it as a means for ensuring traceability, decentralization, and multiparty collaboration by introducing trust in a network of mutually mistrusting entities. This is primarily achieved through consensus mechanisms and distributed ledger architectures, where the degree of ledger replication varies depending on the blockchain type. In public blockchains, full nodes maintain complete ledger copies, while in private and consortium blockchains, often more suitable for VANET deployments, ledger access and replication are restricted to authorized participants

[15, 25]. Researchers of VANETs are no different and are putting in efforts to solve diverse problems using blockchain. However, the fusion of these two apparently relevant technologies gives rise to additional concerns related to feasibility and mutual compliance [15].

4.1. Compliance of Blockchain and VANET

In continuation of the earlier discussion, VANET seems to fit the use case of blockchain just right, but with a catch. VANET is all about real-time communication and the timely delivery of unaltered messages. Mindful thinking about a busy highway, where numerous vehicles hit high speeds simultaneously, covering significant distances within split seconds, depicts how compromising the performance of VANET can cause fatalities. Moreover, the computational burden on the RSUs and OBUs also needs to be feasible despite the recent technical advancements. Blockchain, however, is still not suitable for such highly dynamic, mobile, and computationally constrained applications [15]. It implies that blockchain-based works need improvisation of the existing blockchain architecture to comply with the requirement of VANET [26] [27]. From the same concern, we used three prime aspects for evaluating the compliance of blockchain-based solutions in VANETs as presented in the third table of each of the subcategories of section 5 (e.g., in Section 5.1.1, Table 4) [15].

4.1.1. Mobility

Participants in the blockchain network are called nodes. Trivially, the geographical trajectory of nodes is not something to be concerned about. However, for VANETs, a vehicle node's rapidly changing location and topology of the network must be taken into consideration. This certainly needs to be reflected in the authentication (handover), consensus, and trust evaluation of the proposed schemes [28]. The proposed schemes that do not address this factor will undoubtedly fail in real-world deployment.

4.1.2. Latency

The factor of latency is related to the temporal variance of the VANET. For instance, the alert message for an accident occurring on a highway needs to be propagated rapidly throughout the network for effective traffic control. In addition, an extra delay of a few seconds in the message delivery might even exceed the time validity of the event [26]. The proposed schemes with significant delay are ineffective considering the real-world application.

4.1.3. *Overhead*

Recent advancements in the automotive industry suggest that future cars are going to be far less distinguishable from advanced high-end computers. The OBUs are already burdened with infotainment and assistive features. Exerting it further with something as computationally expensive as blockchain will create a bottleneck in performance. Improvised solutions for blockchain architectures are essential for addressing this aspect [29].

4.2. *Purpose of Using Blockchain in VANET*

Despite the complexities involved, blockchain adoption in VANETs is receiving growing attention from both researchers and automotive industry leaders [30]. This interest stems from blockchain’s potential to foster trust among mutually distrusting entities, introduce elements of decentralization, and enable robust auditability through immutable record-keeping. As detailed in Section 2, our full-text review of the collected papers highlights how VANET researchers are applying blockchain in increasingly nuanced ways. The most prominent approaches are summarized below.

- **Decentralization:** Blockchain is currently regarded as a new means of realizing distributed computing. Considering the vast geographical coverage and large scale of deployment in VANET, decentralization is necessary without a doubt, and blockchain can definitely play a role here [31].
- **Multiparty Collaboration:** Blockchain has the capability of introducing trust among mutually mistrusting entities, leveraging the immutable shared ledger that stores data redundantly to thwart breaches of integrity. This ledger also acts as a single source of truth for auditing in case of anomalies and upholds transparency. In addition, particular scenarios in VANETs require the formation of groups (e.g., group validation of messages), and consortium blockchain is used for managing the groups and secured data sharing among the members [32] [15].
- **Automation:** Some blockchains support smart contracts, which can be executed automatically according to particular business logic. A noticeable number of systems used the smart contracts of blockchain to automate a particular portion of their workflow in a traceable and transparent manner [33].

5. Classification

This section presents a systematic, functionality-based classification of the papers surveyed in our analysis. These works are organized across four primary domains: Authentication (Section 5.1), Trust Management (Section 5.2), Security (Section 5.3), and Privacy (Section 5.4). A key characteristic of the literature is that these domains are not mutually exclusive, as many solutions are inherently multifaceted. This overlap is reflected in the quantitative distribution of classifications across the subsections. The surveyed papers are distributed as follows: Authentication (23 papers), Trust Management (18 papers), Security (13 papers), and Privacy (17 papers), with totals exceeding 55 due to papers addressing multiple domains. Consequently, to provide a complete functional analysis, papers are discussed in each relevant subsection. The respective subcategories under the four domains provide further granularity. Readers should note that the core technical articles surveyed in this review generally correspond to References [26] through [99] in the bibliography.

5.1. Authentication

In the context of VANETs, authentication needs to be achieved at two levels - node level and message level. Initially, a new node (i.e., a vehicle) registers itself to interact in the network as a valid participant. Afterward, the disseminated messages from these vehicles must be verified to determine whether they originated from a valid source or not [34]. From the collected papers, the blockchain-based authentications in VANETs can be grouped into three types, as discussed below.

5.1.1. Authentication of New Vehicles

This subgroup addresses vehicle authentication in VANETs by ensuring valid credentials through trusted intermediaries such as Law Enforcement Authorities (LEAs) or TTPs. These authentication mechanisms inherently intertwine with conditional privacy preservation [3], a relationship further explored in Section 5.4. The surveyed papers in this category are summarized in Table 3 (conceptual overview: contributions, limitations, blockchain architecture) and Table 4 (implementation: tools, metrics, VANET-specific feasibility).

Table 3: Summary of vehicle authentication schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[32]	2019	Consortium	Not re-ported	Multiparty collaboration, Decentralization	Certificateless authentication scheme leveraging edge computing and providing conditional PP	Did not address the group formation in context of VANET
[35]	2019	Consortium	Not re-ported	Automation, Decentralization	PPA framework for automatic authentication using smart contracts and attribute-based encryption, functional even if the TA is offline	Implementation-based evaluation required
[36]	2019	Consortium	Not re-ported	Immutability, Data storage	Proposed a novel Merkle Patricia tree data structure, distributed authentication to replace CRLs, conditional privacy through multiple certificates	Pilot projects for further improvement
[31]	2020	Not re-ported	Not re-ported	Decentralization, Automation, Data sharing	Automated and distributed key management and lightweight authentication through bivariate polynomial key agreement protocol	Blockchain aspect of the scheme remained entirely unaddressed
[37]	2020	Not re-ported	PBFT	Transparency, traceability	Vehicles authenticating using unlinkable pseudonyms leveraging homomorphic encryption	Unlinkability breaks down for cooperation between SMs, and vulnerable to collusion attacks
[38]	2021	Not re-ported	PoW	Automation, Decentralization	Improved version of [39] removing the cloud server and decentralizing authentication and information storage	Implementation done considering a single vehicle and single RSU only
[40]	2021	Not re-ported	Not re-ported	Data storage, Data sharing	Authentication, emergency message storage and distribution protocol	Use of blockchain limited to storing and distributing of EMs
[41]	2021	Consortium	Kafka consensus service	Data storage, Multiparty collaboration	Randomized messages of Epileptic curve cryptosystem for PPA, used asynchronous accumulator for membership verification	Extra layer of RSMs needed for managerial role, increasing jurisdictional complexities
[42]	2021	Not re-ported	Not re-ported	Data storage, Traceability	Biometrics blockchain framework for secured communication and retaining genuine identity of users anonymously	Blockchain used for storage only
[43]	2023	Private	PBFT	Immutability, Transparency	Leveraged feedback of participants for certificate issuance and revocation, used smart contracts for certificate authentication	Evaluation system of feedback was not addressed elaborately

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Table 3: Summary of vehicle authentication schemes – continued from previous page

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[33]	2023	Public	PoW	Multiparty collaboration	Used PKC for initial handshaking, channel phase response for key extraction, mitigated redundant computational cost of authentication	No explicit comparison of cost given between proposed approach and maintaining actual certificates
[44]	2023	Public	PoW	Storage, Transparency	PPA protocol supporting user-controlled conditional tracability, used auxiliary data processors, maintaining valid identities and revocation through blockchain	Scalability and privacy-utility trade-off needs to be addressed
[45]	2023	Not re-ported	Not re-ported	Transparency	TPD-based PPA algorithm using blockchain to store the parameters later used for signature verification	Used blockchain server, not compliant with the claim of decentralization
[46]	2023	Consortium	PBFT	Auditing	PPA that reduces computation cost of message authentication through batch verification and elliptic curve, used smart contracts for forensics	System can be made more lightweight, and fine-grained access control can be introduced
[47]	2024	Consortium	PBFT	Secure Communication	Proposed a dual blockchain system for efficient authentication and communication in VANETs	Future work involves evaluating the system under large-scale VANET scenarios

Vehicle authentication schemes reveal fundamental architectural trade-offs between decentralization, performance, and privacy. Edge-assisted approaches leverage RSU computation [32] or smart contract automation for TTP-independent operation [35], while decentralized credential management eliminates revocation bottlenecks through Merkle Patricia Trees [36] or bi-variate polynomial key agreements [31]. Privacy-preserving schemes achieve unlinkable pseudonyms via homomorphic encryption [37] but introduce collusion vulnerabilities among validators. Hybrid architectures address storage scalability by integrating IPFS with blockchain hash referencing [38, 39], scaling to national emergency distribution [40], or eliminating CRLs through regional signature managers despite jurisdictional complexity [41]. Recent frameworks unify multiple objectives: reputation-based trust with automated identity management [43], certificate replacement via smart contracts [33], conditional traceability [44], dual blockchain architectures separating authentication from communication [47], and batch verification optimized for VANET latency [46]. Critical limitations persist: reliance on tamper-proof devices without addressing sophisticated attacks [45, 42], centralized components contradicting decentralization claims, and inadequate validation under realistic traffic conditions.

Critical Insights and Subtle Pitfalls. A closer technical examination of the surveyed literature reveals several limitations that often remain implicit or only partially addressed.

- A considerable portion of schemes that are presented as decentralized rely heavily on certificate authorities or comparable trusted entities. These authorities typically maintain highly protected centralized cloud infrastructures that manage identities, credentials, and policy enforcement. Blockchain components are then incorporated mainly to provide immutable storage and traceability. Such systems inevitably reintroduce centralized trust and therefore do not satisfy the architectural expectations of decentralized authentication. This situation introduces a fundamental architectural question concerning whether the added operational cost of a blockchain, particularly a public blockchain with significant consensus overhead, is justified in the presence of a centralized trust anchor [45, 40, 42].
- Many proposals assume that tamper-proof devices within on-board units provide consistently strong security guarantees. However, practical evidence indicates that these devices remain vulnerable to cyber

Table 4: Implementation tools, metrics, and feasibility overview of vehicle authentication schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[32]	Not reported	Theoretical analysis, Computation cost, Storage overhead, and Communication cost	✗	✗	✓
[35]	Hyperledger-composer, JS	Time overhead, Run-time cost	✗	✗	✗
[36]	IBM's HLF platform v1.1,	Transaction throughput and latency, Time cost, Communication overhead, Computational cost	✗	✓	✓
[31]	OMNeT++, Veins	computational overhead, packet loss ratio	✗	✓	✓
[37]	SUMO, ProVerif	Computation overhead, Communication overhead, Consensus time	✓	✓	✗
[38]	Ethereum, Remix IDE	Computation cost, Communication cost, storage cost, deploy cost in GAS, transaction cost, execution cost	✗	✗	✗
[40]	Truffle framework, Ganache emulator, Metamask, NPM	Computational Overhead, Storage overhead, Throughput, Average Delay, Packet dropping rate	✗	✗	✓
[41]	Hyperledger Fabric	Throughput, Latency	✓	✓	✓
[42]	SUMO, OM-NeT++	Computation cost, packet loss, packet delay	✗	✓	✗
[43]	SUMO, OM-NeT++, Raspberry Pi	Message rate, message and processing delay	✗	✓	✓
[33]	HLPSL, MIRACL, Ethereum Mainnet, OMNeT++, SUMO	Logical analysis, Communication cost, authentication delay and average packet loss ratio	✗	✓	✓
[44]	Ethereum, Vanet-MobiSim, NS-2	Throhgput, delay	✓	✓	✓
[45]	VanetMobiSim	Average delay	✗	✓	✗
[46]	MIRACL, Hyperledger Fabric	Computational overhead, verification delay	✓	✓	✓
[47]	OMNeT++, SUMO	Authentication delay, success rate, throughput	✓	✓	✓

compromise and targeted physical interference. Any protocol that relies on their perfect integrity consequently inherits an inherently fragile security boundary [45].

- Several works provide only marginal conceptual extensions to prior frameworks. Instead of reexamining core assumptions or addressing fundamental shortcomings of earlier designs, they introduce minor adjustments to parameters or workflow patterns. These modifications rarely alter the threat model or yield measurable performance improvements, and thus contribute limited technical advancement [35, 31].

5.1.2. Message Authentication

Following the initial authentication of vehicles as mentioned earlier in Section 5.1.1, the subsequent challenge is to ensure the integrity of their communications. The VANET’s proper functionality relies on the timely, secure, and authentic dissemination of messages. Message authentication ensures that messages originate from legitimate sources without compromising vehicle privacy [48]. While authentication serves as a prerequisite for trust evaluation, verifying cryptographic origin rather than behavioral trustworthiness, a distinction elaborated in Section 5.2, only a subset of reviewed papers explicitly focus on this foundational aspect. The surveyed papers of this subcategory are summarized in Table 5 (conceptual overview) and Table 6 (implementation and feasibility).

Table 5: Summary of blockchain-based message authentication schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[49]	2019	Not re-ported	Not re-ported	Transparency, Decentralization	Certificate-less public key signature employing batch signature and aggregation signature verification in V2I communicating	Blockchain aspect remained entirely unaddressed
[50]	2021	Not re-ported	Not re-ported	Data storage, Decentralization	Efficient batch signature and aggregation signature verification similar to [49]	Scalability remained unaddressed, implementation required
[51]	2021	Public	PoW	Multiparty collaboration, Transparency	Ring signatures for message verification, blockchain-powered IPFS for solving storage problem in VANET	Ring formation not described amply in context of VANET, the long process of registration of a witness before event validation is not practical in real-life situations
[52]	2022	Not re-ported	Not re-ported	Transparency	A threshold ring signature-based anonymous traffic event validation scheme similar to [51]	Comparative analysis with the SOTA not provided
[46]	2023	Consortium	PBFT	Auditing	Significantly reduced computation cost of message authentication using batch verification and elliptic curve cryptography	Blockchain can be improved for achieving further efficiency
[53]	2023	Public	PoW	Multiparty Collaboration	Blockchain assisted certificate revocation using coded caching, which can quickly verify messages and reduce network load	The blockchain aspect is not compliant with VANET as per the argument of paper [54]
[55]	2023	Public	PoW	Traceability	Secure data sharing through blockchain and traceable ring sharing algorithm	How the rings are going to be formed in a highly mobile network like VANET is left unaddressed
[56]	2024	Consortium	PBFT	Trust computation	Integrated trust computation with privacy-preserving authentication using blockchain	Future work will focus on optimizing trust evaluation for dynamic environments

Table 6: Implementation tools, metrics, and feasibility overview of blockchain-based paper for message authentication schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[49]	MIRACL	Signing cost, Execution cost	✗	✓	✓
[50]	Java JBPC Library	Execution time, computation cost	✗	✗	✗
[51]	Python, Ethereum, Metamask, Remix IDE, Oyente	Time cost, reputation value, gas consumption	✗	✗	✗
[52]	SUMO, OM-NeT++	Delay, time cost, participation rate	✗	✓	✗
[46]	MIRACL, Hyperledger Fabric	Computational overhead, verification delay	✓	✓	✓
[53]	Python-bitcoinlib, PYBC, ML-RPC, OpenSSL	CRL size, Retrieval delay	✓	✓	✓
[55]	PYBC	Computation cost	✗	✗	✓
[56]	Hyperledger Fabric	Authentication time, trust score accuracy	✓	✓	✓

Message authentication schemes prioritize verification efficiency while preserving sender anonymity through cryptographic aggregation and ring-based protocols. Ali et al. [49] employ batch and aggregate signature verification to reduce RSU latency and enable blockchain-based pseudonym revocation, though the blockchain integration remains superficial. Ren et al. [50] similarly leverage blockchain for conditional traceability, revealing identities only upon misbehavior detection. Ring signature approaches achieve stronger anonymity but introduce operational challenges: Khalid et al. [51] combine blockchain-powered IPFS with anonymous ring signatures for traffic event validation, yet their complex registration process risks unacceptable delays, while Ahmed et al. [52] propose threshold ring signatures without comparative evaluation against existing schemes. Zhang et al. [55] extend ring signatures to secure traffic data sharing but leave ring formation mechanisms unspecified for highly mobile environments. Optimization strategies focus on reducing cryptographic overhead: Tao et al. [46] apply batch verification with elliptic curve cryptography (discussed in Section 5.1.1), while Liang et al. [53] introduce BAC-CRL, using coded caching for certificate revocation with reduced network load despite reliance on suboptimal PoW consensus. Rawat et al. [56] integrate these concerns through BTC2PA, unifying trust computation with conditional privacy-preserving authentication in a single blockchain-based framework.

Critical Insights and Common Pitfalls. Some of the critical shortcomings observed from the mentioned works are stated below.

- Group formulation is a necessary aspect for the message authentication schemes but most works do not precisely address how the group membership of vehicles will be managed in the context of their rapidly changing location all the time [49, 52, 55].
- Message authentication in VANET must be done very frequently and in real time. So, it is important to investigate the feasibility of using blockchain technology in this scenario [57].

5.1.3. Handover of Mobile Vehicles

Handover, a fundamental concept in mobile networks, occurs when communicating nodes move from one coverage area to another. In VANETs, where RSUs provide network coverage and vehicles are highly mobile, frequent location changes make repeated authentication impractical [58]. To address this, various blockchain-enabled handover authentication schemes have been proposed. These surveyed papers are summarized in Table 7 (conceptual overview) and Table 8 (implementation and VANET-specific feasibility).

Table 7: Summary of blockchain-based handover authentication schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[59]	2020	Public	PoW	Data storage, Decentralization	Time-efficient handover through generating handover certificates and tokens by the RSUs	Changes in vehicle attributes are recorded in the Bitcoin ledger, incurring high latency
[29]	2022	Consortium	Algorand (PoS + BFT)	Multiparty Collaboration, Data Storage	Lightweight blockchain-based V2I initial authentication and handover authentication protocol reducing unnecessary computations	Implementing the scheme in a real-life scenario and improving the system by reducing the block consensus time to lessen the burden on RSU
[60]	2023	Consortium	Raft	Multiparty Collaboration, Data sharing	Cross-domain PPA using group signature, where blockchain facilitates trusted data sharing, namely user registration and facilitating information query	Improve usability through new blockchain mechanism
[28]	2023	Not reported	Not reported	Integrity	Blockchain-based mutual authentication and session key agreement protocol by utilizing the Elliptic Curve Cryptography (ECC) and one-way hash function	The protocol was investigated informally, and usage of two blockchains requires much work to be done for making the system suitable for real-time handover of high-speed vehicles.
[26]	2023	Not reported	Not reported	Transparency, Data sharing	Lightweight handover scheme leveraging blockchain for transparent data sharing between RSUs for commencing handover, used PUF and biological key for security of RSUs	The system can be improved through improving the blockchain technology
[61]	2024	Hybrid	PBFT	Privacy preservation	Developed a hybrid blockchain-based scheme for privacy-preserving authentication and secure handovers	Future work includes exploring other privacy-preserving mechanisms for further efficiency

Table 8: Implementation tools, metrics, and feasibility overview of blockchain-based paper for handover authentication schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[59]	GNU multiple precision arithmetic (GMP), Pairing-based cryptography (PBC) library	Time cost	✓	✗	✓
[29]	NS3, MIRACL library, PBC Library, Raspberry Pi, AVISPA	Computational cost, end-to-end delay, throughput	✓	✓	✓
[60]	Hyperledger Fabric, SUMO, OM-NeT++	Computational overhead, average delay	✓	✓	✓
[28]	Real-or-Random oracle model, Scyther tool	Communication cost, computation cost	✓	✗	✓
[26]	Theoretical analysis, Raspberry Pi	Authentication time, Storage overhead	✓	✓	✓
[61]	MATLAB, Hyperledger	Authentication delay, handover latency	✓	✓	✓

Handover authentication schemes address the challenge of maintaining security during frequent RSU transitions without repeating full authentication protocols. Approaches diverge between certificate-based and cryptographic efficiency strategies. Wang et al. [59] employ blockchain-stored trust values and handover certificates, but Bitcoin transaction latencies preclude real-time applicability. Son et al. [29] achieve lightweight re-authentication through hash and XOR operations following initial V2I validation, identifying block consensus time (the network-wide validation delay distinct from local cryptographic computation) as the primary overhead bottleneck. Cross-domain scenarios introduce additional complexity: Chen et al. [60] apply group signatures with blockchain-facilitated data sharing for inter-domain handovers, acknowledging unresolved blockchain usability issues. Architectural innovations attempt to balance security with performance: Dwivedi et al. [28] propose dual-ledger systems (auxiliary and parent blockchains) with ECC-based mutual authentication, though this structure requires refinement for high-speed mobility, while Su et al. [61] employ hybrid public-consortium architectures to preserve privacy during handovers. Xie et al. [26] extend coverage beyond RSU range through lightweight protocols enabling pseudonymous V2V communication in infrastructure-sparse regions, critical for emergency

scenarios despite needing blockchain architectural improvements.

Critical Insights and Common Pitfalls. A few observations from the above-mentioned handover schemes are pointed out below.

- The schemes need to address the further granularity of handover scenarios, for example, in overlapping coverage areas and even areas out of coverage or discontinuity [59, 29, 60].
- As handovers will occur frequently, comparing the handover with the initial authentication is not sufficient; the computational feasibility of handover should also be discussed [28].

5.2. Trust Management

For this genre of work, it is assumed that vehicles are correctly authenticated and message authentication mechanisms function as intended within the VANET. However, authentication alone does not guarantee the trustworthiness of network participants. Authenticated vehicles may still exhibit malicious behavior, such as falsely reporting accidents to implicate specific vehicles or deliberately disseminating inaccurate traffic information to gain personal advantage [12]. In this environment of mutually mistrusting entities, researchers leverage blockchain technology to establish fairness, mitigate malicious activities, and foster trust across the network. Based on different approaches, the papers are classified into three categories, as presented below.

5.2.1. Rule-based Trust Evaluation

This subcategory encompasses studies that employ rule-based strategies for trust evaluation in VANETs through the assessment of specific parameters related to message dissemination. These approaches may be entity-centric, data-centric, or hybrid in nature and are designed to quantify trustworthiness within particular contexts or events. Rule-based mechanisms maintain comparatively lower computational overhead and latency than smart contract-based or feedback-driven trust systems (elaborated in Sections 5.2.2 and 5.2.4), rendering them particularly suitable for real-time VANET applications. The surveyed papers in this subcategory, which are discussed individually below to substantiate these characteristics, are summarized in Table 9 (conceptual overview) and Table 10 (implementation and feasibility).

Table 9: Summary of rule-based trust evaluation schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[62]	2019	Not reported	PoW, PoS	Auditing, Decentralization	Security framework for vehicular IoT environment in SDN enabled 5G-VANETs where the blockchain stores traffic information as well as the source of that information to detect any malicious activity	The video report service still uses a centralized approach
[63]	2020	Private	Not reported	Data management, Integrity	Blockchain-based data management, edge-assisted trust evaluation, trusted cloaking areas based on cooperative vehicle trust values, enhances security and reliability while reducing latency in trust assessments	Blockchain architecture of the scheme not amply addressed
[64]	2020	Public	Thermal Reactor	Data storage, Transparency	Adopted custom PBFT protocol for detecting and removing malicious vehicles	Trust incentives can be introduced for encouraging fair behaviours
[59]	2020	Public	PoW	Decentralization	Used blockchain for storing vehicle attributes and trustworthiness immutably and establishing trust	Changes in vehicle attributes are recorded in the Bitcoin ledger, which has a significantly high latency
[51]	2021	Public	PoW	Multiparty collaboration, Transparency	Reputation and incentive mechanism for trust, used blockchain-powered IPFS as storage solution	Membership management and revocation not discussed elaborately
[65]	2021	Consortium	PBFT	Transparency, Immutability	Used consortium blockchain networks for trust score aggregation, enhanced performance by detecting and blacklisting malicious nodes	Proceeded with an impractical assumption on the number of malicious nodes
[66]	2024	Consortium	PBFT	Trust management, Secure data exchange	Proposed a trust mechanism combining blockchain and UAVs in VANETs	Future work will address the scalability of the trust mechanism

Table 10: Implementation tools, metrics, and feasibility overview of rule-based trust evaluation schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[62]	OMNeT++ 4.5, Crypto++ library 5.6.2	Processing time, detection accuracy, transmission delay, video encryption time overhead	✗	✗	✓
[63]	Python	Computation cost, communication time, Probability of location privacy leakage	✗	✓	✓
[64]	Hyperledger fabric	percentage of malicious vehicles, Trust value tendency, delay of constructing anonymous cloaking regions, probability of privacy information leakage	✗	✗	✓
[59]	GNU multiple precision arithmetic (GMP), Pairing-based cryptography (PBC) library	Time cost	✗	✗	✓
[51]	Python, Ethereum, Metamask, Remix IDE, Oyente	Time cost, reputation value, gas consumption	✗	✗	✗
[65]	OSM, SUMO, NS-2, AWK, java environment	Computation cost of transaction pool processing and trust score aggregation, block time	✗	✗	✓
[66]	Not reported	Trust level	✗	✗	✗

Rule-based trust evaluation schemes quantify trustworthiness through parameter-driven scoring without computationally intensive machine learning, trading adaptability for real-time feasibility. Architectural choices reveal persistent centralization-decentralization tensions: Xie et al. [62] store trusted traffic data on-chain while relegating encrypted videos to centralized cloud storage, reintroducing tampering vulnerabilities their blockchain component aims to prevent. Feng et al. [63] mitigate central dependence through TCAC’s integration of consortium blockchain with edge-assisted regional trust evaluation, while Li et al. [64] employ thermal reactor consensus, which is simpler than PBFT, more secure than PoX, to purge vehicles below trust thresholds. Consensus mechanism selection critically impacts real-time viability: Wang et al. [59] calculate trust via trustworthiness attribute information and levels [67] but record updates as Bitcoin transactions with prohibitive latency, while Khalid et al. [51] similarly encounter Ethereum’s un-

suitability for dynamic vehicular environments despite ring signature-based event validation. Alternative validation architectures emerge to address V2V unreliability: Motorcu et al. [66] offload trust rule enforcement to UAVs for off-chain validation, improving accuracy through aerial perspective. Kudva et al. [65] contribute entity-based frameworks with two-level scoring and PBFT consensus for insider threat detection, though their focus on black-hole attacks alone and assumption of less than 25% malicious nodes limit generalizability.

Critical Insights and Common Pitfalls. Despite all the performance benefits of rule-based trust evaluation, some problems need attention, as noted below.

- Rule-based approach can not incorporate the implication of historical trust [64, 66, 68].
- Continuous trust evaluation, storage, and management through the blockchain itself gets computationally burdensome [62, 59].

5.2.2. *Data-Driven Trust Evaluation*

This subcategory comprises studies that employ data-driven approaches for trust evaluation by leveraging the underlying relationships between vehicles and their disseminated messages. These schemes utilize various machine learning algorithms to assess trustworthiness based on behavioral patterns and historical data. The surveyed papers in this subcategory are summarized in Table 11 (conceptual overview) and Table 12 (implementation and feasibility).

Table 11: Summary of blockchain-based data-driven trust schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[69]	2019	Public	PoW, PoS	Trust, Data sharing	Bayesian Inference Model for message validation, generating ratings for message sources, and uploading these ratings to RSUs for calculating trust values	Use of vehicle identity number as an identifier can breach identity privacy
[68]	2019	Consortium	PBFT	Data storage, Integrity	Developed a trust calculation method based on the Dirichlet distribution, taking the location of vehicles into consideration	Scalability issues and assumptions regarding the behavior of malicious vehicles
[70]	2019	Public	PoW, PBFT	Data sharing, Transparency	Logistic regression for enhancing the sensitivity of reputation of vehicles	Lack of sensitivity analysis and discussion regarding simulation parameters
[71]	2020	Not reported	Not reported	Data storage	Node trustability calculated using neural networks, accompanied by rule-based evaluation	Blockchain was only used to store the keys; it could have also been used in trust management
[72]	2021	Consortium	PBFT	Membership management, Data sharing	Hidden Markov Model-based trust evaluation model trained using an unsupervised method and alliance chain for trust management, considering historical behavior of vehicles	Membership management and revocation not discussed elaborately
[73]	2021	Public	PoW, PBFT	Decentralization	Bayesian inference-based validation mechanism for identifying malicious messages	Directly assumed vehicles to be compliant in message dissemination; fault tolerance not addressed
[56]	2024	Consortium	PBFT	Trust computation	Integrated trust computation with privacy-preserving authentication using blockchain	Future work will focus on optimizing trust evaluation for dynamic environments

Table 12: Implementation tools, metrics, and feasibility overview of blockchain-based data-driven trust schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[69]	Matlab	Calculation of trust value offsets(stakes), Block generation time, Transmission latency of messages and ratings	✗	✗	✗
[68]	Not reported	The resilience to attacks, the impact of bad-mouthing attacks, the variation trend of the vehicle's trust level	✓	✓	✓
[70]	Theoretical analysis	Reputation value, Recognition rate of malicious vehicles, average latency	✓	✓	✗
[71]	MATLAB	Number of rejection	✓	✗	✗
[72]	Fabric-iot, Hyperledger Fabric	Time cost, trust evaluation accuracy	✗	✓	✓
[73]	Python	Computation time, block generation time, model precision	✗	✗	✗
[56]	Hyperledger Fabric	Authentication time, trust score accuracy	✓	✓	✓

Data-driven trust evaluation leverages statistical inference and machine learning to assess behavioral patterns, offering adaptability beyond static rule-based systems at the cost of computational complexity. Probabilistic approaches dominate early schemes: Yang et al. [69] apply Bayesian inference to generate vehicle ratings aggregated by RSUs into blockchain-recorded trust offsets using joint PoW and PoS consensus, though real-time feasibility remains unvalidated. Luo et al. [68] similarly employ Dirichlet distributions for RSU-managed trust classification and historical updates but provide no implementation evidence. Liu et al. [70] combine Bayesian trust models with anonymous group signatures in BTCPS, yet omit critical simulation and sensitivity analysis. The necessity of blockchain in data-driven trust remains contentious: Malik et al. [71] argue that hybrid rule-based and ML trust prediction requires only cloud or edge storage for scores, relegating blockchain to key management alone. Advanced machine learning integration emerges in recent work: Liu et al. [72] apply Hidden Markov Models with unsupervised RSU training on Hyperledger Fabric alliance chains for privacy-preserving evaluation, though membership revocation in high-mobility contexts requires refinement. Practical deployment challenges per-

sist across probabilistic methods: Ghajar et al. [73] validate event messages via Bayesian inference incorporating sender distance and reliability but assume unrealistic vehicle collaboration and introduce latency through PoW identity generation. Rawat et al. [56] address these limitations through conditional privacy-preserving systems integrating multi-dimensional real-time and historical data for dynamic trust computation with improved accuracy and reduced latency.

Critical Insights and Common Pitfalls. Data-driven trust evaluation is indeed a prospective aspect of trust management in VANET. However, as noted below, certain aspects of it must be addressed carefully.

- Some works did not utilize the blockchain to enforce trustworthiness. They merely stored keys, identity information, or reputation scores. Using blockchain for mere storage is questionable [69, 71].
- The inference times and real-time performance require comprehensive experimentation and investigation [69, 68, 70].
- Considering the widespread geographical coverage, distributed machine learning, reinforcement learning, and federated learning can prove to be highly effective and practical [71, 72, 56].

5.2.3. Custom Consensus-Based Trust

This subcategory represents a particularly innovative direction in blockchain-enabled VANET trust management. The consensus mechanism constitutes a fundamental component of blockchain technology, serving as the primary means of establishing trust in open, permissionless networks. However, conventional consensus algorithms such as PoW, PoS, and PBFT were not designed to accommodate the stringent real-time and latency-sensitive requirements inherent to vehicular environments [12, 14]. Recognizing this critical gap, researchers have proposed customized consensus mechanisms specifically tailored to VANET constraints, effectively transforming consensus itself into a trust evaluation tool rather than merely a transaction validation mechanism. The surveyed papers in this subcategory are summarized in Table 13 (conceptual overview) and Table 14 (implementation and feasibility).

Table 13: Summary of custom consensus-based trust schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[74]	2019	Not reported	PoE	Transparency, Traceability	Proposed a novel proof-of-event consensus mechanism and a two-pass threshold validation algorithm	Consensus algorithm not vetted appropriately
[75]	2020	Private	Customized PBFT	Transparency, Consensus	Voting-based customized PBFT consensus using blockchain, and efficient relay node selection algorithm integrated into the consensus	Update frequency of the records stored in the blockchain not addressed
[76]	2021	Not reported	Proof of Trust (PoT), PoS	Data sharing	Proposed a trusted routing scheme using FT-OLSR protocol and proof-of-trust (PoT) consensus mechanism	Revocation needs further addressing to be compliant with fault tolerance
[77]	2023	Consortium	Custom algorithm based on DPoS and PoW	Multiparty Collaboration, Transparency	Novel hierarchical and trust certificate service; proposed a new block structure and custom consensus protocol based on DPoS and PoW	The distance factor can be included, which may improve practicality
[78]	2024	Consortium	PBFT	Content delivery, Trust management	Designed a blockchain-based trusted content delivery mechanism for VANET	Future work includes enhancing scalability for real-time applications

Table 14: Implementation tools, metrics, and feasibility overview of custom consensus-based trust schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[74]	NS-3	event success rate, false event success rate	✓	✓	✓
[75]	OMNeT++, SUMO	Hop time, delay, success of dieminated message	✓	✓	✓
[76]	NS-3	Resource consumption and overhead, block generation time, detection time of attackers	✓	✓	✓
[77]	Hyperledger Fabric	Time cost	✓	✓	✓
[78]	Hyperledger, NS-3	RContent delivery success rate, trust accuracy	✓	✓	✓

Custom consensus mechanisms repurpose blockchain validation for trust evaluation itself, transforming consensus from a transaction ordering tool into a behavioral assessment framework tailored to VANET constraints. Event-driven approaches validate traffic incidents rather than generic transactions: Yang et al. [74] introduce Proof-of-Event (PoE) where RSUs perform two-phase validation before irreversible blockchain recording, though mathematical rigor and reliance on potentially compromised OBU tamper-proof devices undermine feasibility. Economic incentives guide validator selection in Ayaz et al.’s [75] voting-based consensus, which incorporates price and reputation to elect relay nodes but suffers from high block generation latency, unspecified update frequencies, and absent privacy protections. Trust itself becomes the consensus criterion in advanced schemes: Inedjaren et al. [76] integrate Proof-of-Trust (PoT) with fuzzy logic routing (FT-OLSR) for blackhole detection, yet lack revocation mechanisms for behavioral changes and incur prohibitive overhead. Hierarchical architectures attempt to balance security with performance: Zhu et al. [77] combine DPoS, PoW, and trust certificate services in HCSC monitored via blockchain-enabled RSUs, though spatial proximity integration could enhance practical applicability. Bibi et al. [78] synthesize these concepts in TR-Block, evaluating node behavior and reliability scores directly within consensus for trust-centric content delivery, achieving high integrity and delivery success in simulation but requiring validation under adversarial conditions.

Critical Insights and Common Pitfalls. Without a doubt, the works on customized consensus algorithms have significant potential, but designing them is much more complex than intuitively enforcing a set of rules. This is

elaborately discussed in section 6.5.

5.2.4. *Feedback-Based Trust Evaluation*

This subcategory encompasses one of the earliest and most widely adopted approaches in blockchain-enabled VANET trust management. Feedback-centric schemes operate by enabling vehicles to provide context-aware evaluations of peer behavior, from which trust scores are computed and aggregated. These mechanisms typically incorporate economic incentive structures wherein participants expend credits or tokens when submitting feedback, thereby deterring malicious or false reports, while rewarding honest contributions to encourage active participation. This dual mechanism of cost imposition and reward distribution renders feedback-based approaches particularly well-suited for the dynamic and mutually mistrusting vehicular environment. The surveyed papers in this subcategory, which are discussed individually below to substantiate these operational characteristics, are summarized in Table 15 (conceptual overview) and Table 16 (implementation and feasibility).

Table 15: Summary of feedback-based trust evaluation schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[79]	2018	Public	BFT	Anonymity, Incentive	An efficient anonymous vehicular announcement aggregation protocol providing witness-based feedback	Use of vehicle identity number as an identifier can breach identity privacy
[80]	2018	Public	PoW	Data storage	Preventing the spread of forged messages through reputation evaluation, neighbour vehicles can directly submit evidence against malicious vehicles	The incurred delay for evidence collection and arbitration for updating reputation score must be addressed robustly for real-time communication
[52]	2022	Not reported	Not reported	Transparency	Threshold Ring signature-based anonymous traffic event validation scheme allowing participants to anonymously witness the message's reliability	Did not compare their work against the existing ones
[43]	2023	Private	PBFT	Immutability, Transparency	Time-aware blockchain-based reputation management leveraging feedback of participants for certificate issuance and revocation	Reputation evaluation system needs elaboration
[81]	2023	Consortium	PoA	Decentralization, Transparency	Decentralized trust management scheme that robustly analyzes and stores vehicle attributes for decision making and revoking certificates, local danger warning dissemination for rapidly notifying the surrounding vehicles	Requires extensive analysis on fault tolerance due to the assumptions of the scheme
[82]	2024	Consortium	PBFT	Trust management, Incentives	Introduced incentive-based decentralized trust management for secure VANET communication	Future research will include testing under diverse traffic scenarios

Table 16: Implementation tools, metrics, and feasibility overview of feedback-based trust evaluation schemes

Paper	Tools used		Metrics Used	Mobility	Latency	Overhead
[79]	NS-2, Java		Total time of announcements, transaction part for users, and consensus part for RSUs	✓	✓	✗
[80]	Python		Reputation score, Storage overhead, and time consumption	✗	✗	✗
[52]	SUMO, NeT++	OM-	Delay, time cost, participation rate	✗	✓	✗
[43]	SUMO, NeT++, Raspberry Pi	OM-Rasp-	Message rate, message delay, and processing delay	✗	✓	✓
[81]	SUMO, NeT++	OM-	Reputation rate, false negative rate	✓	✓	✓
[82]	OMNeT++		Trust accuracy, incentive effectiveness	✓	✓	✓

Feedback-based trust evaluation employs witness attestations and economic incentives to assess participant reliability, introducing game-theoretic considerations absent from rule-based systems. Economic models anchor early schemes: Li et al. [79] introduce CreditCoin where reputation coins reward message forwarding, though incomplete participant motivation and key management weaknesses limit adoption. Evidence accumulation approaches face inherent latency challenges: Lu et al.’s BARS model [83, 80] logs direct and indirect interactions on blockchain to prevent fake messages but cannot mitigate delays from evidence collection and arbitration processes critical for real-time safety applications. Incremental extensions offer limited novelty: Ahmed et al. [52] apply threshold ring signatures building on prior work [51] without substantive innovation or comparative evaluation, while Li et al. [43] propose time-aware feedback-based reputation management but omit benchmarking against established frameworks [83, 75, 51, 65]. Regional architectures attempt to localize trust management: Fernandes et al. [81] integrate voting-based feedback with certificate revocation and local danger warnings in lightweight regional blockchains, yet this fragmentation raises concerns about inter-region handover smoothness and attack resilience. Han et al. [82] synthesize these concepts through dynamic incentive frameworks where blockchain immutably records feedback while adapting trust scores and rewards to network and traffic conditions, demonstrating improved security and information sharing.

Critical Insights and Common Pitfalls. Nonetheless, there are some key points on which future research work can be improved.

- Witness-based feedback-centric schemes are susceptible to bottlenecks since the feedback is dependent on the response of other vehicles [79, 80, 52].
- Schemes must be tested extensively to make sure not to miss out on any special scenarios that might later be used to breach the trust [52, 81].
- The validation and voting feedback mechanism must be compliant with the highly mobile environment of VANET [81, 43, 82].

5.3. Security

As with other cyber-physical systems, VANETs are vulnerable to a spectrum of security threats. This subsection examines studies that leverage blockchain-based approaches specifically to mitigate these threats. While the authentication and trust management mechanisms discussed in Sections 5.1 and 5.2 inherently contribute to network security, the works reviewed here are distinct in their primary focus on security enhancement. These solutions are classified into the following categories.

5.3.1. Secured Data Transmission

This subcategory addresses the critical challenge of secure data transmission in VANETs, where the open wireless medium and highly dynamic topology create substantial vulnerabilities. The broadcast nature of vehicular communication, combined with the presence of numerous mobile nodes entering and leaving the network, exposes transmitted data to interception, tampering, and unauthorized access. To mitigate these risks, researchers have leveraged blockchain’s inherent integrity-preserving and tamper-proof characteristics to ensure secure and verifiable data transmission. The surveyed papers in this subcategory are summarized in Table 17 (conceptual overview) and Table 18 (implementation and feasibility).

Table 17: Summary of blockchain-based secured data transmission schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[84]	2019	Consortium	PBFT	Data storage, Data sharing	Secured data sharing and storage system using asymmetric encryption and signature verification techniques	Implementation not properly addressed
[85]	2020	Public	PoW, PoL	Decentralization, Transparency	A geographically constrained blockchain for secured message passing	The proposed work is not established through experimental analysis
[86]	2020	Not reported	Not reported	Access control, Automation	A secure data sharing architecture using blockchain, IPFS, and CP-ABE encryption method	The retrieval function of VANET data on blockchain and user attribute revocation function not implemented
[71]	2020	Not reported	Not reported	Data storage	Used data sanitization for data security where the required keys of sanitization are stored in blockchain	Blockchain was not amply utilized, rather used as a storage solution
[55]	2023	Public	PoW	Traceability	Secure data sharing through blockchain and traceable ring sharing algorithm; used smart contract to trace and report malicious vehicles	Ring formation in dynamic and mobile VANET environment not addressed
[87]	2024	Not reported	Not reported	Data integrity	Introduces a Bayesian DAG blockchain with edge computing for secure and scalable data transmission	Future work involves optimizing the Bayesian inference model for high-density traffic scenarios
[88]	2024	Not reported	Not reported	Data sharing, preservation	Combines blockchain and zero-knowledge proofs (ZKP) for secure, anonymous authentication in VANETs	Future work involves reducing ZKP computation time for better real-time performance

Table 18: Implementation tools, metrics, and feasibility overview of blockchain-based secured data transmission

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[84]	Not reported	computational delay, communication overhead, verification delay, transmission overhead	✗	✓	✗
[85]	Not reported	Not reported	✓	✗	✓
[86]	Ethereum, Docker	Encryption consumption, storage cost, file transmission time	✗	✗	✓
[71]	MATLAB	Number of rejection	✓	✗	✗
[55]	PYBC	Computation cost	✗	✗	✓
[87]	OMNET++, SUMO	Throughput, latency, scalability	✓	✓	✓
[88]	Not specified	Authentication delay, privacy preservation	✓	✓	✓

Secure data transmission schemes leverage blockchain’s immutability to protect vehicular communications, though many implementations utilize blockchain primarily for storage rather than exploiting its consensus and decentralization properties. Consortium architectures dominate early approaches: Zhang et al. [84] propose DSSCB with digital signatures and PBFT consensus where RSUs store blockchain replicas to distribute storage risk, yet lack implementation details precluding reproducibility. Geographic constraints offer theoretical alternatives: Shrestha et al. [85] introduce localized blockchains where miners generate blocks from event messages for regional sharing, but provide no empirical validation. Access control mechanisms integrate cryptographic sophistication: Li et al. [86] combine attribute-based encryption (HECP-ABE) with blockchain for fine-grained VANET data access, though retrieval efficiency requires optimization. The blockchain-as-storage paradigm raises justification concerns: Malik et al. [71] employ metaheuristic algorithms to tune sanitization keys distributed via blockchain across RSUs, yet this limited usage may not warrant blockchain overhead. Ring signature schemes face mobility challenges: Zhang et al. [55] apply blockchain with ring signatures for secure sharing but leave ring formation mechanisms unspecified for dynamic vehicular topologies. Advanced architectures emerge to address scalability: Alkhalifa et al. [87] introduce DAG-based Bayesian frameworks with edge-assisted RSUs and probabilistic verification for improved throughput and latency. Zhang et al. [88] integrate privacy preservation through Zero-Knowledge Proofs with blockchain for anonymous authentication and

tamper-proof communication achieving low latency.

Critical Insights and Common Pitfalls. The implications of the above-mentioned works still need to be reconsidered from the perspective of the following comments.

- Almost none of the works could establish a compelling link between blockchain and enhancing security and resilience against attacks. Most of them used blockchain for immutable storage and non-repudiation [84, 71, 86].
- Some works lack implementation. Even if it is there, the experimental setup was not adequately comprehensive or practical, posing a question about the reproducibility of those works [85, 84, 86].

5.3.2. Attack Prevention

This subcategory encompasses studies that specifically leverage blockchain technology to detect, prevent, and mitigate common attack vectors in VANETs. Vehicular networks are inherently vulnerable to various malicious activities, including Sybil attacks, replay attacks, message forgery, and denial-of-service attempts, which can compromise both safety-critical communications and network integrity. Researchers have explored how blockchain’s decentralized validation, immutability, and consensus mechanisms can serve as defensive mechanisms against these threats. The surveyed papers in this subcategory are summarized in Table 19 (conceptual overview) and Table 20 (implementation and feasibility).

Table 19: Summary of blockchain-based attack prevention schemes

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[89]	2019	Public	PoW	Multiparty collaboration	Derived the necessary design conditions for a regional blockchain in VANET to minimize the 51% attack	The starting depth of the regional blockchain is kept as future work
[27]	2021	Not reported	PoD	Multiparty collaboration	Proposed a standard service score to narrow down potential miner selection pool and executed PBFT on the reduced pool that is more secured	An attacker can hack the system by being honest for a long time and then becoming a miner node when it has gained enough trust of the system depending on the increased probability for it to be a miner node
[90]	2022	Not reported	Custom PoS	Transparency	Security scheme for malicious node detection and reputation-based detection of tampered messages using a custom incentive-based PoS consensus	Session key used for vehicles is not addressed properly
[91]	2023	Private	Not reported	Decentralization	Blockchain-based signature scheme where the message signature and encryption occur simultaneously for computational efficiency	Testing is needed in multiple OBU's to check the scalability of the scheme
[26]	2023	Not reported	Not reported	Transparency, Data sharing	Used physically unclonable function (PUF) against RSU-capture attacks and biological key for security of RSUs	The system can be improved by improvising the blockchain for VANET
[92]	2024	Not reported	Not reported	Intrusion logging	Develops a blockchain-secured federated learning system for intrusion detection in VANETs	Future work focuses on reducing communication overhead in federated learning

Table 20: Implementation tools, metrics, and feasibility overview of blockchain-based attack prevention schemes

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[89]	Bitcoin, NS-3	Attack success probability	✗	✗	✗
[27]	Java SE	Number of nodes	✓	✗	✓
[90]	SUMO, ONE	False alarm rate, Missed detection rate, Delivery Rate, and Overhead Rate	✗	✗	✓
[91]	MIRACL	Computational cost, communication overhead	✗	✗	✓
[26]	Theoretical analysis, Raspberry Pie	Authentication time, storage overhead	✓	✓	✓
[92]	Ethereum, MetaMask	Detection accuracy, communication overhead	✓	✓	✓

Attack prevention schemes employ blockchain to defend against both vehicular-specific and blockchain-native threats, yet often introduce new vulnerabilities through the integration itself. Theoretical analysis addresses consensus vulnerabilities: Shrestha et al. [89] derive design parameters limiting 51% attack success probability based on node composition, message delivery timing, and computational puzzle difficulty. Consensus refinement attempts to resist network-layer attacks: Kudva et al. [27] introduce Proof-of-Driving (PoD), filtering PBFT miners by service scores to counter MAC layer attacks, though this enables strategic adversaries to exploit gradual trust accumulation for later compromise. Architectural separation isolates attack surfaces: Liu et al. [90] employ dual blockchains with RSUs managing BCIR and CAs managing BCCA to detect selfish behavior and message tampering, yet assume high-end hardware inappropriate for resource-constrained vehicular environments. Cryptographic integration reduces protocol complexity: Shao et al. [91] propose certificateless signcryption unifying encryption and signature generation to eliminate single points of failure and reduce V2V overhead, though bilinear pairing removal and realistic scenario testing remain necessary. Physical security extends beyond cryptographic defenses: Xie et al. [26] augment handover protocols (discussed in Section 5.1.3) with Physically Unclonable Functions and biological keys protecting RSUs from physical capture. Collaborative detection leverages distributed learning: Mansouri et al. [92] integrate blockchain with federated learning for privacy-preserving intrusion detection, maintaining model integrity and transparency while identifying DoS and spoofing attacks with high accuracy.

Critical Insights and Common Pitfalls. Some of the key limitations

of these works are noted below.

- Using blockchain as a source of truth inevitably introduces a new dimension of threat related to exploiting the weaknesses of the blockchain technology itself [89, 27, 90].
- Additionally, making the blockchain VANET compliant opens new vulnerabilities. Prevention against this also requires attention [27, 89].
- It can be argued that very few works have prevented security attacks by using the blockchain itself. Rather, they used other provisions and tools (e.g., PUF, signcryption, etc.) to achieve the goal [91, 26, 92].

5.4. *Privacy*

As established in Section 1, privacy preservation constitutes a fundamental requirement for practical VANET deployment. This subsection examines works that employ blockchain specifically for privacy preservation, despite the inherent tension between blockchain’s transparency and privacy requirements, particularly in public blockchain architectures. Researchers have addressed this challenge through diverse strategies: some leverage blockchain as an immutable ledger while implementing privacy through complementary cryptographic techniques, while others adopt consortium blockchains that provide inherent data visibility control. While numerous works in Sections 5.1, 5.2, and 5.3 claim to offer privacy preservation, this subsection focuses exclusively on studies where privacy preservation represents the primary contribution. The collected papers are organized into the following subcategories.

5.4.1. *Encryption*

This subcategory encompasses studies that employ encryption techniques to maintain the confidentiality of sensitive identity information and critical message content in blockchain-enabled VANETs. Encryption serves as a fundamental privacy-preserving mechanism, preventing unauthorized access to vehicular data that could otherwise compromise user anonymity and location privacy [3]. The surveyed papers in this subcategory are summarized in Table 21 (conceptual overview) and Table 22 (implementation and feasibility).

Table 21: Summary of blockchain-based privacy preservation schemes for VANET using encryption

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[35]	2019	Consortium	Not reported	Automation, Decentralization	Attribute-based encryption (ABE) was enforced and three-factor security to protect privacy	No simulation-based results
[37]	2020	Not reported	PBFT	Transparency, Traceability	Homomorphic encryption used for unlinkability of pseudonyms	Vulnerable to collusion attack
[93]	2021	Not reported	Raft	Data sharing, Transparency	Employed a decentralized deep learning model and used fully homomorphic encryption to secure the parameters of the model, input data, and also its output	Further improvement is possible in blockchain
[94]	2022	Not reported	PoW	Transparency	Privacy-preserving data query service leveraging proxy-based file encryption in keyword searching	Further contribution transmission and computational constraints
[95]	2024	Not reported	DAG	Data integrity, Privacy preservation	Integrates homomorphic encryption with DAG blockchain for secure and private data computation in VANETs	Future work focuses on improving computational efficiency for real-time applications

Table 22: Implementation tools, metrics, and feasibility overview of blockchain-based privacy preservation schemes for VANET using encryption

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[35]	Hyperledger-composer, JavaScript	Time overhead of cryptographic operations, Runtime cost of Smart Contracts	✗	✗	✗
[37]	SUMO, ProVerif	computation overhead, communication overhead, consensus time	✓	✓	✗
[93]	NS-3, SUMO	average communication time, communication overhead, accuracy	✓	✓	✓
[94]	SUMO, NS-3	Delay, gas consumption	✗	✓	✗
[95]	BigChainDB	Computation time, privacy level, scalability	✓	✓	✓

Encryption-based privacy preservation protects sensitive identity and message content through cryptographic transformations, with blockchain providing auditability rather than confidentiality itself. Attribute-based approaches delegate key management to trusted authorities: Feng et al.’s BPAS (Section 5.1.1) employs attribute-based encryption with three-factor security where TTPs issue attribute secret keys [35], while Liu et al. apply homomorphic encryption enabling vehicles to generate multiple unlinkable pseudonyms without central coordination [37]. Computational offloading addresses performance constraints: Chen et al. shift processing from cloud to edge computing nodes, encrypting traffic condition information via Fully Homomorphic Encryption (FHE) for secure parameter updates distributed over blockchain-based ledgers [93]. Query privacy extends to data retrieval: Yeh et al. introduce BPSDQS combining blockchain and IPFS for transparency with proxy-based encryption supporting privacy-preserving keyword searches, though blockchain contributes transparency rather than privacy itself [94]. Advanced frameworks integrate encryption with alternative ledger structures: Jiang et al. combine homomorphic encryption with DAG blockchains enabling secure computation on encrypted data with scalable, tamper-resistant validation for high-traffic scenarios [95]. These schemes reveal a common pattern where blockchain serves audit and integrity functions while cryptographic mechanisms provide actual confidentiality, raising questions about blockchain’s necessity when simpler append-only logs might suffice.

Critical Insights and Common Pitfalls. The reviewed works primarily leverage blockchain for limited purposes, leaving several critical aspects

unaddressed, as noted below.

- The works primarily exploit blockchain for its auditing capabilities and transparent data sharing properties, without leveraging blockchain’s broader potential, such as decentralized consensus mechanisms or smart contract automation [35, 94, 95].
- None of these studies rigorously evaluated whether the performance overhead introduced by blockchain technology justifies its adoption in real-time vehicular communication scenarios, leaving a significant gap in assessing practical feasibility for latency-sensitive VANET applications [35, 37, 93, 94].

5.4.2. Privacy Preservation Through Groups

This subcategory comprises studies that employ group-based anonymity mechanisms to preserve vehicular privacy. The majority of works implement anonymous aggregate vehicular announcement protocols, wherein vehicles are clustered based on contextual attributes such as location, destination, or route. These schemes enable anonymous operations within group boundaries, preventing adversaries from identifying individual vehicles while permitting aggregate group-level tracking for legitimate purposes. This approach effectively balances privacy preservation with accountability requirements in VANETs. The surveyed papers in this subcategory are summarized in Table 23 (conceptual overview) and Table 24 (implementation and feasibility).

Table 23: Summary of blockchain-based privacy preservation schemes for VANET using groups

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[79]	2018	Public	BFT	Anonymity, Incentive	Anonymous vehicular announcement aggregation protocol allowing vehicles to announce anonymously through non-deterministic signers generating signatures	Used static vehicle identity number, which may cause identity privacy breach
[70]	2019	Public	PoW, PBFT	Data sharing, Transparency	Used identity-based group signature mechanism for achieving identity privacy	Sensitivity analysis and simulation parameters need further disclosure
[32]	2019	Consortium	Not reported	Multiparty collaboration, Decentralization	Used Chinese Remainder Theorem for efficient group key updating to facilitate privacy	The group formation is not compliant with high mobility scenarios
[96]	2019	Public	PoW	Transparency, Automation	Leveraged high-dimensional polynomials and sub-identities for IPP while forming k-anonymous groups for uploading SBMs in a group	Latency of the blockchain network remained unaddressed
[52]	2022	Not reported	Not reported	Transparency	Used ring signature for event validation while ensuring privacy of validator vehicles	Weak comparative analysis
[44]	2023	Public	PoW	Storage, Transparency	Used Pointcheval-Sanders (PS) signature, Signature of Knowledge (SOK), and Zero-Knowledge Proofs (ZKPOK) for privacy preservation in VANET	Scalability and privacy-utility trade-off
[61]	2024	Hybrid	PBFT	Authentication, Group privacy	Combines hybrid blockchain with group-based privacy-preserving authentication in VANETs	Plans to explore adaptive group management for enhanced scalability

Table 24: Implementation tools, metrics, and feasibility overview of blockchain-based privacy preservation schemes for VANET using groups

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[79]	NS-2, Java	Total time of announcements, transaction part for users, and consensus part for RSUs	✓	✓	✗
[70]	Theoretical analysis	Reputation value, Recognition rate of malicious vehicles, average latency	✓	✓	✗
[32]	Not reported	Theoretical analysis, Computation cost, Storage overhead, and Communication cost	✗	✗	✓
[96]	OPNET, Ethereum	System time, Probability of location leakage	✗	✗	✓
[44]	Ethereum, Vanet-MobiSim, NS-2	Throhgput, delay	✓	✓	✓
[61]	Hyperledger	Authentication delay, privacy level	✓	✓	✓

Group-based privacy preservation achieves anonymity by obscuring individual identities within collectives, balancing unlinkability with accountability through cryptographic aggregation. Witness-based validation schemes face identifier vulnerabilities: Li et al. (Section 5.2.4) enable anonymous announcement propagation through signatures but compromise privacy when static vehicle identifiers persist across sessions [79]. Group signature mechanisms provide stronger anonymity: Liu et al.’s BTCPS (Section 5.2.2) employs identity-based group signatures for anonymous announcement aggregation [70], while the V2R authentication scheme (Section 5.1.1) applies the Chinese Remainder Theorem within consortium blockchains for dynamic V2V group key updates despite requiring group management refinement. Sub-identity approaches further conceal real identities: Li et al. use threshold secret sharing to distribute identity verification across multiple sub-identities [96]. Hybrid architectures combine blockchain types for flexibility: Su et al. integrate public and private blockchains where group signatures authenticate vehicles without individual revelation, enhancing scalability [61]. Advanced cryptographic protocols strengthen privacy guarantees: the PPA protocol (Section 5.1.1) combines Pointcheval-Sanders signatures, signatures of knowledge, and zero-knowledge proofs for user privacy protection, acknowledging unresolved scalability and privacy-utility trade-offs [44].

Critical Insights and Common Pitfalls. This genre of privacy-

preservation works was reasonably available in the VANET domain before the advent of blockchain. Incorporating blockchain in these works must not neglect the following comments in order to realize a practical solution to it.

- Group management is mostly done without the direct use of blockchain. Rather, they use different existing tools and mechanisms [79, 32, 96, 52].
- Consortium blockchain can be taken as a promising solution, but its highly dynamic topology makes its use questionable [32, 61].

5.4.3. Location Privacy

This subcategory surveys schemes for location privacy preservation in VANETs, a critical concern as spatial data can be used for adversary tracking. To resolve the inherent conflict between blockchain’s transparency and location privacy, researchers have integrated blockchain with complementary privacy-preserving techniques. These approaches aim to protect vehicular location data while maintaining decentralized trust. The surveyed papers in this subcategory are summarized in Table 25 (conceptual overview) and Table 26 (implementation and feasibility).

Table 25: Summary of blockchain-based location privacy preservation in VANET

Paper	Year	Blockchain Type	Consensus	Function of BC	Contribution	Limitation or Future Work
[68]	2019	Consortium	PBFT	Data storage, Integrity	Ensured location privacy preservation (LPP) employing distributed k-anonymity and construction of a trusted anonymous cloaking region	Implementation required for establishing the claims
[96]	2019	Public	PoW	Transparency, Automation	Anonymous uploads of SBMs and maintaining location privacy using an undirected graph generation algorithm	Latency of the blockchain network remained unaddressed
[63]	2019	Consortium	Not reported	Data management, Integrity	Selected cooperative vehicles from neighboring vehicular regions based on trust values to establish a trusted cloaking area	Blockchain architecture of the scheme not addressed clearly
[64]	2020	Public	Thermal Reactor	Data storage, Transparency	Employed the RSUs to accumulate query requests from the vehicles and build an anonymous cloaking area, delivered to the location service provider	Incentive models can be used to improve the cooperation of vehicles
[97]	2024	Public	PoW	Trust management, Data integrity	Develops a blockchain-based trust management system for preserving location privacy in VANETs	Future work includes optimizing pseudonym management for dynamic environments
[98]	2024	Public	PoW	Data integrity	Combines federated learning with blockchain for privacy-aware location prediction in IoV	Evaluation limited to computational performance and completely lacks network simulation

Table 26: Implementation tools, metrics, and feasibility overview of blockchain-based location privacy preservation in VANET

Paper	Tools used	Metrics Used	Mobility	Latency	Overhead
[68]	Not reported	The resilience to attacks, the impact of bad-mouthing attacks, the variation trend of the vehicle's trust level	✓	✓	✓
[96]	OPNET, Ethereum	System time, Probability of location leakage	✗	✗	✓
[63]	Python	Computation cost, communication time, Probability of location privacy leakage	✗	✓	✓
[64]	Hyperledger fabric	percentage of malicious vehicles, Trust value tendency, delay of constructing anonymous cloaking regions, probability of privacy information leakage	✗	✗	✓
[97]	Ethereum, Python	Location privacy, trust accuracy	✓	✓	✓
[98]	DESIGN EXPERT V13	Prediction accuracy, communication overhead	✗	✗	✗

Location privacy preservation employs spatial obfuscation techniques to prevent trajectory inference while maintaining service utility, with blockchain primarily providing audit trails rather than privacy mechanisms themselves. Cloaking region approaches aggregate vehicles into anonymity sets: Luo et al. construct trust-anonymous cloaking regions through distributed k-anonymity with novel trust data structures, though lacking implementation validation [68], while Li et al. apply graph-based k-anonymity ensuring location diversity within groups to prevent similarity-based de-anonymization [96]. Trust integration guides participant selection: Feng et al.'s TCAC (Section 5.2.1) forms trusted cloaking areas by selecting cooperative vehicles based on trust scores [63]. Architectural separation isolates privacy-critical operations: Li et al. employ dual blockchains where CerBC manages certifications and ReqBC handles anonymized requests, enabling traceability for anomalies without compromising routine location privacy [64]. Extended frameworks unify multiple privacy techniques: Li et al. combine pseudonym-based cryptography with blockchain-maintained immutable trust records for location obfuscation [97]. Federated learning offers alternative privacy paradigms: Ali et al. predict locations without raw data sharing, using blockchain to en-

sure model integrity and distributed trust with demonstrated accuracy and privacy protection [98].

Critical Insights and Common Pitfalls. Taking a step back, the justification for using blockchain for location privacy was not amply conveyed in most of the works. In accordance with that, researchers on this topic should find the following comments useful.

- Blockchain inherently does not play any role in the preservation of location privacy other than auditing in case of an anomaly [68, 96, 63, 64].
- Most of the LPP schemes work with anonymous cloaking regions [68, 63, 64].
- Focusing on fundamental algorithmic ideas (e.g., graph generation [96]) to improve existing LPP schemes can be more impactful.

5.4.4. *Pseudonymity*

A common approach to conditional privacy in VANETs involves allowing vehicles to use pseudonyms, digital disguises that conceal real identities during network interactions. While many papers mention pseudonym usage, only a few provide detailed frameworks for blockchain-based pseudonym management.

Pseudonymity management enables conditional privacy by allowing identity revelation for accountability while preserving routine anonymity, though blockchain's high throughput demands for frequent pseudonym operations raise performance concerns. Dual blockchain architectures attempt to separate lifecycle phases: Benarous et al. propose a "blockchain of blockchains" model where one public permissionless ledger stores vehicle-managed certified pseudonyms while another public permissioned ledger maintains RSU-managed revocations, yet this approach lacks implementation evidence and incurs overhead from maintaining parallel ledgers [99]. Merkle tree structures offer cryptographic efficiency: Ren et al. (Section 5.1.2) employ dual Merkle trees for storing and revoking self-generated pseudonyms but omit blockchain type specifications critical for evaluating consensus latency [50]. Cryptographic generation enables unlimited pseudonyms: the BUA framework (Section 5.4.1) applies homomorphic encryption allowing vehicles to generate pseudonyms validated by consortium blockchain service managers using locally stored encrypted data, avoiding centralized pseudonym distribution

bottlenecks [37]. These schemes reveal tension between pseudonym change frequency required for unlinkability and blockchain throughput limitations, with some authentication systems deliberately decoupling pseudonym management from blockchain to maintain real-time performance.

Critical Insights and Common Pitfalls. Considering the small number of papers, no tabular representation for this type of privacy preservation model is provided herein. Nonetheless, some important research questions need a solution for this subcategory, which is discussed below.

- Pseudonym management requires a high amount of throughput for both reading and writing of the information; using blockchain for this task inherently raises performance questions. Some blockchain-based authentication systems exist that do not associate the pseudonym management with blockchain to make sure no delay gets incurred, which might hamper the operation of VANET [45].
- Some works suggest storing pseudonyms in blockchain to maintain transparency, but that raises the question of confidentiality, requiring additional workload for encryption or any other necessary means [37].
- A general trend is to use pseudonyms in PPA schemes merely, but attackers can link multiple pseudo-identities to obtain the successive location of vehicles [100].

6. Challenges and Discussions

The systematic review of blockchain-based solutions in Section 5 revealed promising approaches for addressing VANET challenges. However, the integration of these two technologies faces significant obstacles that are often inadequately addressed in current research. This section examines the key challenges observed across authentication, trust management, security, and privacy domains, along with a critical analysis of current approaches and their limitations.

6.1. Authentication Challenges

Authentication forms the foundation of secure vehicular communication, yet blockchain integration introduces specific complexities that current research has not adequately resolved.

Regulatory Compliance vs. Decentralization. The fundamental tension between blockchain’s decentralized nature and VANET’s regulatory requirements remains unresolved. While blockchain promises distributed trust, vehicular networks operate under strict oversight from transportation authorities, law enforcement, and regulatory bodies. This creates an inherent conflict: authentication systems must simultaneously support autonomous vehicle-to-vehicle verification while maintaining centralized accountability for legal and safety compliance. Current proposals often ignore this duality, presenting either fully decentralized solutions that lack regulatory hooks or centralized systems that merely use blockchain as a redundant storage layer.

Performance Degradation in High-Mobility Scenarios. Authentication in VANETs demands sub-second response times, particularly during high-speed encounters or dense traffic conditions. However, blockchain consensus mechanisms introduce latencies ranging from seconds to minutes, creating a fundamental mismatch with vehicular requirements. The computational overhead of cryptographic operations, combined with block propagation delays, can transform authentication from a security enabler into a performance bottleneck. Studies consistently overlook these timing constraints, presenting theoretical models without empirical validation under realistic mobility patterns.

State Synchronization Complexity. The authentication process requires a consistent state across multiple system layers: the blockchain ledger, vehicle certificates, roadside infrastructure databases, and authority registries. Maintaining synchronization across these heterogeneous systems while vehicles rapidly transition between network domains presents an unsolved challenge. Without robust synchronization protocols, authentication decisions may rely on outdated or conflicting information, potentially compromising both security and safety. The lack of standardized interfaces between blockchain and existing vehicular PKI systems exacerbates this problem.

Scalability of Certificate Management. Vehicle authentication typically involves certificate verification, revocation checking, and pseudonym management. Storing and processing these operations on-chain quickly becomes infeasible as the network scales. For instance, a metropolitan area with 100,000 vehicles generating hourly pseudonym changes would produce millions of transactions daily, overwhelming current blockchain capacities. While some proposals suggest off-chain certificate storage with on-chain hashes, this approach reintroduces centralized dependencies and negates many of

blockchain’s purported benefits.

Cross-Domain Authentication Failures. Vehicles frequently traverse multiple administrative domains, requiring seamless authentication handovers between different blockchain networks or consensus groups. Current research lacks practical solutions for inter-blockchain authentication, particularly when domains operate different consensus protocols or have conflicting trust assumptions. This gap becomes critical at borders, between service providers, or when transitioning between urban and highway environments with different infrastructure operators.

Potential Solution Directions. The authentication challenges identified above require coordinated research efforts across multiple dimensions. The tension between regulatory compliance and decentralization (Section 7.1, Section 7.7) necessitates hybrid architectures that maintain audit trails while enabling peer-to-peer verification. Performance degradation in high-mobility scenarios demands adaptive consensus mechanisms and lightweight cryptographic protocols (Section 7.5, RQ5.1-5.3). State synchronization complexity can be addressed through co-design strategies that harmonize blockchain protocols with vehicular network characteristics (Section 7.2, RQ2.1-2.3). Certificate management scalability benefits from partition-tolerant consensus and hybrid ledger architectures (Section 7.2, RQ2.2). Cross-domain authentication requires standardized interfaces and interoperability protocols (Section 7.8, RQ8.1-8.2). These solution directions are explored in detail in Section 7 with concrete, testable research questions.

6.2. Trust Management Challenges

Trust management in blockchain-based VANETs faces fundamental design flaws that compromise both theoretical soundness and practical applicability.

Computational Infeasibility of Reputation Systems. Most of the blockchain-based trust proposals rely on complex reputation calculations involving historical behavior analysis, multi-party voting, and weighted scoring algorithms. These computations must occur in real-time for every vehicle interaction, creating an impossible computational burden. For example, verifying trust scores for 50 nearby vehicles, each requiring consensus among 10 witnesses with 100 historical transactions, would demand processing thousands of blockchain queries within milliseconds. This computational explosion renders reputation-based systems impractical for real-world deployment.

The Witness Reliability Paradox. Trust calculation often depends on witness vehicles to validate behavior and events. However, this creates a circular dependency: determining witness trustworthiness requires other witnesses, leading to infinite regression or arbitrary trust bootstrapping. Moreover, in sparse traffic conditions, insufficient witnesses may be available, while in dense scenarios, coordinating among numerous witnesses introduces prohibitive communication overhead. The dynamic nature of vehicular networks, where witnesses constantly change, further undermines the stability of witness-based validation.

Incentive Mechanism Failures. Economic incentive models borrowed from cryptocurrency systems fail to capture the complexity of vehicular interactions. Unlike financial transactions with clear monetary value, vehicular behaviors involve multidimensional factors including safety, efficiency, comfort, and regulatory compliance. Reducing these to scalar reward values oversimplifies decision-making and can create perverse incentives. For instance, rewarding vehicles for reporting incidents might encourage false reports, while penalizing contested reports might discourage legitimate warnings.

Machine Learning Integration Obstacles. While AI and reinforcement learning approaches promise adaptive trust evaluation, their integration with blockchain faces severe practical limitations. Training models requires extensive historical data that may not exist for new vehicles or rare events. The computational requirements for running neural networks on resource-constrained vehicular hardware conflict with real-time processing needs. Furthermore, the black-box nature of many ML models contradicts blockchain’s transparency principles, making trust decisions unauditible and legally problematic.

Trust Transferability Issues. Trust established in one context may not transfer to another. A vehicle’s reputation for accurate traffic reporting doesn’t necessarily indicate trustworthiness in emergency situations. Current schemes lack granular trust models that differentiate between various interaction types, temporal factors, and contextual conditions. This one-size-fits-all approach to trust can lead to security vulnerabilities where malicious actors exploit reputation in one domain to conduct attacks in another.

Potential Solution Directions. Trust management challenges demand fundamental rethinking of computational approaches and validation architectures. Computational infeasibility can be mitigated through lightweight protocols and integrated simulation frameworks that identify performance bottlenecks before deployment (Section 7.5, Section 7.4, RQ4.1-4.2, RQ5.2).

The witness reliability paradox requires alternative validation architectures, potentially leveraging infrastructure nodes or federated learning approaches that distribute trust computation (Section 7.6, RQ6.3). Incentive mechanism design benefits from principled justification methodologies that align blockchain properties with vehicular behavior complexity (Section 7.1, RQ1.2). Machine learning integration obstacles can be addressed through privacy-preserving distributed learning frameworks (Section 7.6, RQ6.1). Trust transferability issues require context-aware trust models validated through comprehensive testbed experimentation (Section 7.4, RQ4.3).

6.3. Security Challenges

Security in blockchain-integrated VANETs involves complex interactions between multiple attack surfaces, creating vulnerabilities that current research inadequately addresses.

Consensus Vulnerability Under Mobility. Traditional blockchain consensus assumes relatively stable network membership and connectivity. In VANETs, rapid topology changes can fragment the network, creating isolated consensus groups that may diverge. An attacker controlling vehicles at strategic positions could partition the network, causing consensus splits that enable double-spending or conflicting transaction histories. Regional blockchain approaches that limit consensus scope to improve performance inadvertently amplify these vulnerabilities by reducing the number of nodes an attacker must compromise.

Attack Surface Expansion. Rather than enhancing security, blockchain integration often expands the attack surface. Vehicles must now defend against both traditional VANET attacks (jamming, spoofing, Sybil) and blockchain-specific threats (51% attacks, selfish mining, eclipse attacks). The interaction between these attack vectors creates novel exploitation opportunities. For instance, an attacker might use vehicular Sybil identities to influence blockchain consensus, or exploit blockchain delays to conduct replay attacks on time-sensitive vehicular messages.

Cryptographic Overhead vs. Safety. The extensive cryptographic operations required for blockchain participation, signature generation, verification, hashing, and proof computation consume computational resources that might otherwise support safety-critical functions. In emergency scenarios, vehicles face an impossible choice between maintaining blockchain security and responding to immediate safety threats. This resource competition

is particularly acute in older or lower-end vehicles with limited processing capabilities.

Immutability Complications. Blockchain’s immutability, while preventing tampering, creates problems for security incident response. Malicious transactions or compromised data cannot be removed, potentially propagating false information indefinitely. In safety-critical scenarios where incorrect data could cause accidents, the inability to quickly revoke or correct blockchain entries poses unacceptable risks. Current proposals for mutable blockchains or redaction mechanisms undermine the very properties that make blockchain attractive for security.

Key Management Complexity. Blockchain systems require vehicles to manage cryptographic keys for transaction signing, which introduces severe practical challenges. Key loss renders vehicles unable to participate in the network, while key compromise enables attackers to impersonate vehicles. The frequent pseudonym changes required for privacy multiply key management complexity. Hardware security modules provide partial solutions but add cost and potential failure points. The lack of practical key recovery mechanisms that maintain security while enabling legitimate access remains unsolved.

Potential Solution Directions. Security challenges in blockchain-integrated VANETs require proactive threat modeling and adaptive defense mechanisms. Consensus vulnerability under mobility can be addressed through partition-tolerant designs and adaptive quorum mechanisms that maintain consistency despite network fragmentation (Section 7.2, Section 7.5, RQ2.1, RQ5.1). Attack surface expansion necessitates principled evaluation of whether blockchain integration provides net security benefits over simpler alternatives (Section 7.1, RQ1.1, RQ1.3). Cryptographic overhead versus safety trade-offs demand lightweight security protocols and co-designed architectures that prioritize safety-critical operations (Section 7.5, RQ5.2). Immutability complications can be partially addressed through governance frameworks that balance data retention requirements with correction capabilities (Section 7.7, RQ7.1, RQ7.3). Key management complexity benefits from standardized interfaces and hardware security integration (Section 7.8, RQ8.2). Formal verification methods can detect vulnerabilities before deployment (Section 7.6, RQ6.3).

6.4. Privacy Challenges

Privacy preservation in blockchain-based VANETs faces fundamental contradictions between transparency requirements and confidentiality needs.

The Transparency-Privacy Paradox. Blockchain’s value proposition includes transparent, auditable transactions, yet VANETs require strong privacy protection for user locations and identities. This fundamental contradiction cannot be resolved through cryptographic patches alone. Even with encryption or pseudonymization, the public availability of transaction patterns enables sophisticated correlation attacks. Traffic analysis can reveal movement patterns, social relationships, and personal habits despite cryptographic protections.

Linkability Despite Pseudonymization. Pseudonym schemes aim to prevent long-term tracking by frequently changing vehicle identifiers. However, blockchain’s permanent record enables adversaries to link pseudonyms through transaction analysis, timing correlations, and behavioral patterns. The more frequently pseudonyms change, the more blockchain transactions are generated, creating more data for correlation attacks. This creates an impossible tradeoff between privacy and scalability.

Regulatory Compliance Conflicts. Privacy regulations like GDPR grant users rights to data erasure and correction, which directly conflict with blockchain’s immutability. A user cannot truly delete their vehicular data once recorded on-chain, potentially creating legal liability for system operators. Proposals for off-chain storage with on-chain references only partially address this issue, as the references themselves may contain identifying information or enable reconstruction of private data.

Location Privacy Impossibility. Vehicular applications inherently involve location-dependent services, requiring vehicles to reveal approximate positions for traffic management, toll collection, or emergency services. Recording these interactions on a transparent blockchain, even with obfuscation techniques, provides adversaries with sufficient information for trajectory reconstruction. Differential privacy and k-anonymity approaches that work in centralized systems break down when applied to blockchain’s distributed, replicated structure.

Group Privacy Vulnerabilities. Vehicles often travel in groups (families, fleets, platoons), creating privacy dependencies where one vehicle’s exposed identity can reveal information about others. Blockchain systems that record vehicle interactions or group memberships enable group tracking even

when individual vehicles maintain pseudonymity. The social graph revealed through blockchain transactions provides rich information for adversarial inference.

Potential Solution Directions. Privacy challenges arise from fundamental contradictions between blockchain transparency and confidentiality requirements, demanding innovative cryptographic and architectural solutions. The transparency-privacy paradox can be partially resolved through selective disclosure mechanisms, differential privacy overlays, and hybrid architectures that separate public audit trails from private transaction details (Section 7.6, RQ6.1). Linkability despite pseudonymization requires sophisticated unlinkability techniques validated through adversarial testing (Section 7.4, RQ4.3). Regulatory compliance conflicts necessitate redactable blockchain designs and governance frameworks that reconcile immutability with data protection rights (Section 7.7, RQ7.1-7.3). Location privacy and group privacy vulnerabilities demand principled evaluation of whether blockchain’s audit benefits justify privacy risks, potentially favoring simpler append-only logs for specific use cases (Section 7.1, RQ1.1). Cross-disciplinary collaboration with legal scholars and policymakers is essential for developing acceptable privacy-preserving solutions (Section 7.7, Section 7.8).

6.5. Custom Consensus Protocol Limitations

The development of VANET-specific consensus mechanisms faces inherent tradeoffs that current research has not successfully resolved.

The CAP Theorem Constraint. Custom consensus protocols cannot escape the fundamental CAP theorem limitations: they must sacrifice either consistency, availability, or partition tolerance. In VANETs, network partitions are common due to mobility and environmental obstacles, making partition tolerance essential. This forces protocols to choose between consistency (all nodes see the same state) and availability (the system remains operational). Safety applications require both, creating an impossible requirement that no consensus protocol can fully satisfy.

Scalability-Security Tradeoffs. Reducing consensus participants to improve latency weakens security by lowering the attack threshold. Increasing participants for security degrades performance beyond acceptable limits. Geographic or hierarchical consensus schemes that limit participation to nearby vehicles improve efficiency but enable localized attacks where adversaries concentrate malicious nodes in specific areas. No proposed protocol successfully balances these competing demands under realistic threat models.

Dynamic Membership Complexity. VANET consensus must handle constantly changing participant sets as vehicles enter and leave the communication range. Traditional blockchain consensus assumes relatively stable membership, with well-defined processes for nodes joining or leaving. The rapid, unpredictable membership changes in VANETs can cause consensus protocols to fail or produce inconsistent results. Solutions like view-change protocols or dynamic committee selection introduce additional latency and complexity.

Message Complexity Explosion. Achieving Byzantine fault tolerance typically requires $O(n^2)$ message exchanges, where n is the number of participants. In dense traffic with hundreds of vehicles, this creates unsustainable communication overhead. Proposals for reducing message complexity through probabilistic or optimistic approaches sacrifice fault tolerance, while hierarchical schemes introduce centralization points that become attractive attack targets.

Potential Solution Directions. Custom consensus protocol development requires acknowledging fundamental theoretical constraints while pursuing practical optimizations within these bounds. The CAP theorem constraint necessitates explicit design choices favoring partition tolerance and availability for safety applications, accepting eventual consistency rather than strong consistency (Section 7.2, RQ2.1). Scalability-security trade-offs can be managed through adaptive mechanisms that dynamically adjust participant counts based on network conditions and threat levels (Section 7.5, RQ5.1, RQ5.3). Dynamic membership complexity requires view-change protocols validated through integrated simulation under realistic mobility patterns (Section 7.4, RQ4.2). Message complexity explosion can be mitigated through hierarchical or geo-fenced architectures, accepting centralization trade-offs where appropriate (Section 7.2, Section 7.5, RQ2.1, RQ5.3). Transparent reporting of protocol limitations through reproducible research practices enables informed adoption decisions (Section 7.3, RQ3.1-3.3).

6.6. Problems yet to be solved

Beyond domain-specific issues, the integration of blockchain with VANETs faces systemic challenges that span multiple technical and operational dimensions. Table 27 summarizes these persistent challenges that current research has not adequately addressed.

These challenges reveal a fundamental mismatch between blockchain’s design assumptions and VANET operational requirements. Static, computation-

Table 27: Problems yet to be solved in Blockchain-based VANETs

Problem Area	Specific Challenge	Research Implications
Group Management	Dynamic group management in highly mobile environments	Develop adaptive authentication and privacy preservation mechanisms for constantly changing vehicle groups
Performance	Real-time transaction processing	Create lightweight blockchain solutions that can handle quick, efficient communication in VANETs
Scalability	Handling large volumes of transactions	Design blockchain architectures that can scale with potentially hundreds or thousands of vehicles
Privacy-Utility Balance	Trade-off between location privacy and service efficiency	Develop nuanced approaches that protect individual privacy without compromising network functionality
Security	Vulnerability of blockchain technology	Create robust security protocols that can resist attacks specific to both blockchain and VANET environments
Technology Adaptation	Potential new vulnerabilities when adapting blockchain	Conduct comprehensive security assessments to identify and mitigate risks introduced by blockchain integration
Identity Management	Pseudonym management challenges	Design efficient, privacy-preserving pseudonym generation and rotation mechanisms
Location Privacy	Preserving user location confidentiality	Develop innovative techniques to obscure location data while maintaining network utility
Trust Evaluation	Data-driven trust assessment	Create real-time, privacy-preserving machine learning approaches for trust calculation
Consensus Mechanisms	Custom protocol development	Design lightweight, efficient, and secure consensus protocols tailored to VANET's dynamic nature

intensive consensus mechanisms conflict with dynamic, resource-constrained vehicular environments. The immutable, transparent ledger contradicts privacy and correction needs. Decentralized governance clashes with regulatory oversight requirements.

Section 7 addresses these systemic challenges through eight interconnected research directions. Rather than treating these as isolated optimization problems, we propose coordinated solutions: principled justification methodologies for determining when blockchain provides advantages over alternatives (Section 7.1), co-design strategies harmonizing blockchain with VANET constraints (Section 7.2), reproducible practices enabling cumulative progress (Section 7.3), integrated simulation and testbed validation (Section 7.4), adaptive consensus and lightweight protocols (Section 7.5), proactive resilience mechanisms (Section 7.6), ethical and legal frameworks

Table 28: Challenges–Solutions–RQ Mapping

Challenge (Sec. 6)	Solution Direction (Sec. 7)	RQs
Auth. Compliance	Principled Eval.; Legal/Ethics	RQ1.1–1.3, RQ7.1–7.3
Auth. Performance	Adaptive Consensus; Co-Design	RQ5.1–5.3, RQ2.1–2.3
Auth. State Sync	Co-Design; Standardization	RQ2.1–2.3, RQ8.1–8.2
Auth. Cert. Scalability	Co-Design	RQ2.2
Auth. Cross-Domain	Standardization	RQ8.1–8.2
Trust Computation Cost	Lightweight Prot.; Simulation	RQ5.2, RQ4.1–4.2
Trust Witness Reliability	Resilience; Testbeds	RQ6.3, RQ4.3
Trust Incentives	Principled Justification	RQ1.2
Trust + ML	Privacy-Preserving ML	RQ6.1
Trust Transferability	Comprehensive Testing	RQ4.3
Consensus Attacks	Partition-Tolerant; Adaptive	RQ2.1, RQ5.1
Attack Surface	Principled Evaluation	RQ1.1, RQ1.3
Crypto Overhead	Lightweight Protocols	RQ5.2
Immutability Conflicts	Governance Frameworks	RQ7.1, RQ7.3
Key Management	Standardization; Verification	RQ8.2, RQ6.3
Privacy–Transparency	Privacy-Preserving; Principles	RQ6.1, RQ1.1
Privacy Linkability	Adversarial Testing	RQ4.3
Privacy Compliance	Legal Frameworks	RQ7.1–7.3
Location/Group Privacy	Alt. Assessment; Governance	RQ1.1, RQ7.2
CAP Trade-offs	Design Trade-offs	RQ2.1
Scalability vs Security	Adaptive Mechanisms	RQ5.1, RQ5.3
Dynamic Membership	Simulation Validation	RQ4.2
Message Complexity	Hierarchical; Geo-Fencing	RQ2.1, RQ5.3
Reproducibility	Open Science	RQ3.1–3.3

(Section 7.7), and standardization efforts (Section 7.8). Table 28 provides explicit mappings between identified challenges and proposed solutions with corresponding research questions.

The field must move beyond technological evangelism toward evidence-based engineering that prioritizes system requirements over architectural preferences. The research questions in Section 7 provide concrete, measurable targets for addressing the challenges identified in this section, enabling systematic progress toward practical blockchain-VANET integration.

7. Future Directions

This section synthesizes insights from Section 6 and outlines actionable recommendations for advancing blockchain-VANET integration. The challenges identified across authentication (Section 6.1), trust management (Section 6.2), security (Section 6.3), privacy (Section 6.4), and consensus protocols (Section 6.5) reveal fundamental mismatches between blockchain’s design assumptions and VANET operational requirements. While the reviewed

literature has produced important theoretical foundations, it remains fragmented, with limited validation in operational vehicular environments.

We propose eight interconnected research directions that directly address these challenges through rigorous system design, comprehensive experimentation, and interdisciplinary collaboration. Table 28 provides explicit mappings between Section 6 challenges and the solutions proposed in this section. Each subsection identifies concrete, testable research questions with quantifiable success criteria, enabling systematic progress toward practical blockchain-VANET deployments.

7.1. Principled Justification: Establishing When Blockchain is Appropriate

A consistent theme across the surveyed literature is the tendency to adopt blockchain without critically evaluating its necessity relative to other technologies. Many proposals assume that decentralization and immutability are inherently beneficial, overlooking whether these properties truly address VANET-specific requirements. Future research should begin by articulating clear, context-driven justifications for choosing blockchain over centralized or hybrid trust models. This includes comparative threat modeling to demonstrate why distributed ledgers outperform trusted authorities or federated databases under realistic attack scenarios. For example, in cooperative driving applications, the risk of collusion among infrastructure nodes may indeed warrant decentralized verification, whereas time-critical emergency communication may not.

These questions directly address the regulatory compliance versus decentralization tension (Section 6.1), attack surface expansion concerns (Section 6.3), and transparency-privacy paradoxes (Section 6.4) by establishing empirical thresholds for when blockchain provides measurable advantages.

Key Research Questions:

- *RQ1.1:* Under what node compromise rates (e.g., 10%, 30%, 50% of RSUs) does blockchain-based consensus outperform centralized PKI in terms of attack detection latency and false positive rates?
- *RQ1.2:* What is the quantitative threshold of infrastructure collusion risk at which decentralized verification becomes cost-effective compared to hierarchical trust models in platooning scenarios?
- *RQ1.3:* How do blockchain-based approaches compare against federated databases for misbehavior detection across metrics of detection

accuracy, communication overhead, and convergence time under varying vehicle densities (50-500 vehicles/km)?

7.2. Co-Design for Efficiency: Ensuring Compatibility Between Blockchain and Vehicular Networks

Once the rationale for blockchain adoption is established, research must focus on co-design strategies that harmonize blockchain protocols with the unique performance envelope of VANETs. Existing consensus mechanisms and data structures often fail to accommodate the rapid topology changes and intermittent connectivity typical of vehicular networks. Promising directions include partition-tolerant consensus schemes that maintain local consistency within road segments while deferring global reconciliation to infrastructure nodes. Additionally, hybrid architectures that combine lightweight on-vehicle ledgers with cloud-based aggregation services can balance responsiveness with scalability. Figure 4a underscores progress in addressing mobility, latency, and overhead, but further work is needed to quantify the trade-offs inherent in these designs.

These co-design strategies address performance degradation in high-mobility scenarios, state synchronization complexity, and certificate management scalability (Section 6.1), consensus vulnerability under mobility (Section 6.3), and CAP theorem constraints (Section 6.5).

Key Research Questions:

- *RQ2.1:* What is the optimal partition size (in terms of geographic radius or vehicle count) for segment-based consensus that minimizes global reconciliation overhead while maintaining safety message delivery within 100ms?
- *RQ2.2:* How does hybrid ledger placement (vehicle-edge-cloud) affect transaction finality time and energy consumption across urban, highway, and rural scenarios with vehicle speeds ranging from 30-120 km/h?
- *RQ2.3:* Can adaptive block size mechanisms reduce end-to-end latency by more than 30% compared to fixed-size blocks under dynamic traffic conditions, and what is the impact on storage requirements?

7.3. Transparent, Reproducible, and Iterative Research Practices

The lack of reproducibility in blockchain-VANET studies significantly limits the field's progress. As noted in Section 2, many papers do not disclose

experimental configurations, simulation parameters, or implementation details. This opacity prevents independent verification and impedes comparative evaluation. Future contributions should adopt open science practices by releasing simulation code, datasets, and benchmark scenarios. Researchers can facilitate reproducibility by developing standardized evaluation frameworks that enable side-by-side comparisons of different protocols under identical conditions. Additionally, iterative studies, where previous work is systematically extended through parameter tuning, scenario diversification, or security stress-testing, can accelerate cumulative knowledge.

Addressing the reproducibility crisis directly supports solutions to all challenge areas (Section 6) by enabling rigorous validation of claimed performance improvements, security guarantees, and privacy protections.

Key Research Questions:

- *RQ3.1*: Can a standardized benchmark suite with defined traffic patterns, attack models, and performance metrics reduce result variance across independent implementations of the same protocol to within 15%?
- *RQ3.2*: What minimal set of parameters (e.g., vehicle density, communication range, block interval, consensus quorum) must be standardized to enable meaningful cross-study comparisons?
- *RQ3.3*: How do variations in cryptographic primitives (e.g., ECDSA vs. BLS signatures, SHA-256 vs. SHA-3) affect throughput and latency across five representative blockchain-VANET protocols under identical test conditions?

7.4. Integrated Simulation and Realistic Testbeds

Simulation remains the dominant approach for prototyping blockchain-VANET systems, yet most current evaluations rely on siloed models that separately assess vehicular mobility and blockchain behavior. This separation neglects critical feedback loops, such as the impact of block propagation delays on safety-critical message delivery. To address this gap, researchers should develop integrated simulation platforms that combine vehicular network simulators (e.g., SUMO, Veins) with blockchain frameworks (e.g., Hyperledger Fabric, Quorum, Ethereum clients). Such integration allows holistic examination of consensus latency, transaction throughput, and communication reliability under realistic conditions. Figure 4b shows the variety of tools currently in use, highlighting the need for cohesive frameworks.

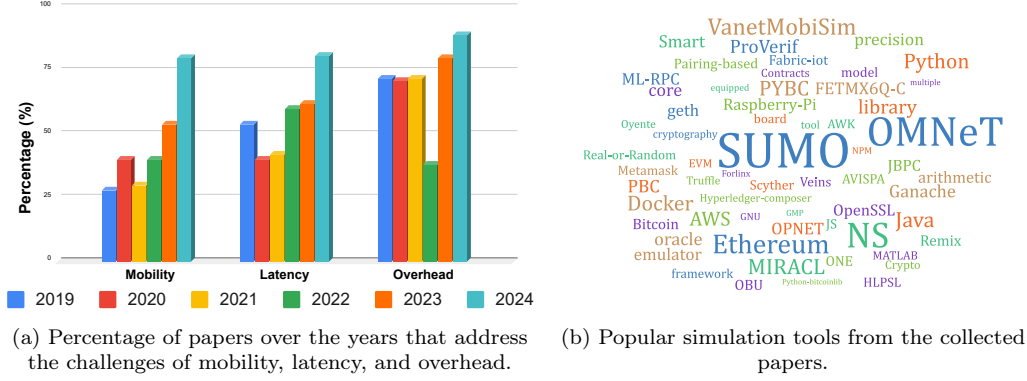


Figure 4: Overview of challenges and simulation tools identified in the collected papers.

Integrated testing addresses witness reliability paradoxes and trust transferability issues (Section 6.2), validates cryptographic overhead impacts on safety (Section 6.3), and enables adversarial testing of linkability attacks (Section 6.4).

Key Research Questions:

- *RQ4.1:* How much does block propagation delay degrade collision avoidance effectiveness (measured by reaction time and accident rate) when blockchain verification is inserted into the V2V safety message pipeline?
- *RQ4.2:* What is the correlation between consensus finality time and packet delivery ratio in integrated simulations versus decoupled approaches across network loads of 100-1000 transactions/second?
- *RQ4.3:* Can field testbed deployments with 10-50 connected vehicles validate simulation predictions within a 20% margin for key metrics (throughput, latency, packet loss) under comparable traffic scenarios?

7.5. Adaptive Consensus and Lightweight Security Protocols

Consensus remains one of the most significant performance bottlenecks in blockchain-VANET integration. Classical protocols like PBFT or PoW are fundamentally misaligned with high-mobility, latency-sensitive environments. Future work should prioritize the development of adaptive consensus mechanisms that dynamically adjust quorum requirements and validation

intervals based on network density, link stability, and transaction urgency. For example, geo-fenced consensus schemes could restrict validation to vehicles within a defined spatial boundary, while deferring global commitment to infrastructure nodes. Simultaneously, lightweight security protocols that minimize cryptographic overhead are essential for maintaining responsiveness.

Adaptive consensus mechanisms directly address performance degradation (Section 6.1), computational infeasibility of reputation systems (Section 6.2), cryptographic overhead versus safety trade-offs (Section 6.3), and scalability-security conflicts (Section 6.5).

Key Research Questions:

- *RQ5.1:* Can adaptive quorum mechanisms that adjust validator count based on network density (e.g., 3-10 validators) reduce consensus latency by 40% while maintaining Byzantine fault tolerance for up to $f < n/3$ malicious nodes?
- *RQ5.2:* What is the trade-off curve between security level (measured by attack success probability) and computational overhead (CPU cycles, energy consumption) for ephemeral pseudonym schemes versus traditional PKI across transaction rates of 10-1000 tx/s?
- *RQ5.3:* How do geo-fenced consensus boundaries (100m, 500m, 1km radii) affect the balance between local responsiveness and global consistency, measured by finality time and fork rate under vehicle speeds of 60-120 km/h?

7.6. Resilience Against Emerging Threats and Privacy Risks

Beyond performance and scalability, future blockchain-VANET solutions must proactively address evolving security and privacy challenges. Adversaries may exploit blockchain transparency to perform trajectory inference attacks, reconstructing vehicle movement patterns despite pseudonymization. Researchers should explore privacy-preserving techniques such as differential privacy overlays, mix networks, or selective disclosure mechanisms that limit data granularity without compromising accountability. Likewise, novel attacks such as consensus eclipse, selfish mining, or validator bribery may manifest in vehicular settings where node churn and intermittent connectivity are pervasive.

These proactive measures address machine learning integration obstacles and witness reliability (Section 6.2), key management complexity and formal verification needs (Section 6.3), and transparency-privacy paradoxes requiring differential privacy (Section 6.4).

Key Research Questions:

- *RQ6.1:* What level of differential privacy noise (ϵ values between 0.1-10) prevents trajectory reconstruction with $>90\%$ confidence while maintaining usable location accuracy (within 50m) for traffic management applications?
- *RQ6.2:* Under what conditions of node churn (vehicles joining/leaving per minute) and network partition (segment isolation duration) do eclipse attacks succeed in isolating validator nodes, and how effective are mitigation strategies like randomized peer selection?
- *RQ6.3:* Can formal verification methods detect consensus vulnerabilities (e.g., liveness violations, safety breaches) in adaptive blockchain-VANET protocols, and what percentage of known attack vectors can be eliminated through pre-deployment verification?

7.7. Ethical, Legal, and Societal Considerations

The integration of blockchain into vehicular systems raises ethical and regulatory questions that must not be overlooked. Immutable ledgers may conflict with data protection regulations like GDPR, which guarantee the right to erasure and informed consent. Moreover, pervasive recording of vehicular interactions could enable surveillance and profiling, undermining user trust and societal acceptance. Future research should engage legal scholars, policymakers, and ethicists to develop governance frameworks that reconcile technological capabilities with privacy rights and accountability obligations.

Governance frameworks address regulatory compliance versus decentralization tensions (Section 6.1), immutability complications for incident response (Section 6.3), and regulatory compliance conflicts with GDPR (Section 6.4).

Key Research Questions:

- *RQ7.1:* Can redactable blockchain designs that support selective data erasure maintain sufficient immutability for accident liability determination while achieving GDPR compliance, measured by audit trail integrity and erasure completeness?

- *RQ7.2:* What governance models (centralized authority, distributed voting, hybrid approaches) achieve the optimal balance between accountability and privacy, evaluated through user acceptance studies with 500+ participants across diverse demographics?
- *RQ7.3:* How do different data retention policies (7 days, 30 days, 1 year) affect the trade-off between forensic utility for incident investigation and privacy risk, quantified by re-identification probability and storage costs?

7.8. Cross-Domain Collaboration and Standardization Efforts

Finally, the future of blockchain-VANET integration depends on cross-domain collaboration and the development of standardized interfaces and protocols. As research matures, aligning efforts across academia, industry consortia, and standards bodies will be vital to achieving interoperability and adoption. Initiatives such as IEEE 1609 (WAVE) and ETSI ITS standards should incorporate guidance for blockchain compatibility, defining clear specifications for message formats, consensus protocols, and security primitives.

[Standardization efforts address cross-domain authentication failures \(Section 6.1\)](#), [key management complexity through unified interfaces \(Section 6.3\)](#), and [enable reproducible research \(Section 7.3\)](#).

Key Research Questions:

- *RQ8.1:* What minimal set of API specifications and message format standards would enable interoperability across three major blockchain platforms (Hyperledger, Ethereum, Quorum) in VANET deployments, verified through cross-platform integration testing?
- *RQ8.2:* Can backward-compatible extensions to IEEE 1609.2 accommodate blockchain-based certificate management without exceeding current latency budgets (100ms for safety messages)?
- *RQ8.3:* What are the quantifiable benefits (in terms of development time, deployment cost, and adoption rate) of standardized reference implementations compared to proprietary solutions, measured through industry case studies and pilot programs?

In summary, the path forward demands a shift from isolated, proof-of-concept solutions toward rigorous, reproducible, and context-aware designs.

The research questions outlined above provide concrete targets for empirical investigation, enabling systematic progress in blockchain-VANET integration. By embracing principled justification, co-design for performance, reproducible practices, integrated testbeds, adaptive consensus protocols, proactive security measures, and cross-disciplinary governance, researchers can transform blockchain from a conceptual novelty into a dependable enabler of intelligent transportation systems. Sustained commitment to these objectives will be essential to realize the promise of secure, efficient, and privacy-preserving blockchain-VANET deployments.

8. Conclusion

This paper provides a comprehensive review of blockchain-based solutions for authentication, trust management, security, and privacy preservation in VANETs. Through methodical classification of the literature, it offers newcomers a clear and structured overview of this domain. The tabular representations and detailed analysis of limitations provide a qualitative understanding of current trends and practices, enhancing the survey’s clarity and utility.

One key contribution is the intuitive categorization of existing research, which simplifies the learning curve for those entering the field. By grouping similar approaches and highlighting overlaps, the paper presents a nuanced perspective on how various blockchain solutions address different VANET challenges. This approach clarifies the current state of research while underscoring the diversity and complexity of applying blockchain technology in vehicular networks.

The paper also examines how blockchain’s fundamental properties, immutability, decentralization, transparency, and cryptographic security, address specific VANET requirements. For instance, immutability ensures tamper-resistant storage of vehicle credentials and traffic data, while decentralization eliminates single points of failure in trust management. However, these same properties introduce trade-offs: immutability complicates the revocation of compromised credentials, decentralization increases consensus overhead in high-mobility scenarios, and transparency conflicts with privacy requirements. The open discussion on challenges and future directions directly connects to these properties. Scalability issues stem from blockchain’s consensus mechanisms struggling with VANET’s dynamic topology and high transaction volumes. Privacy concerns arise from blockchain’s inherent trans-

parency requiring additional cryptographic layers. Interoperability challenges reflect the tension between decentralized architectures and existing centralized transportation infrastructure.

Future research must address these property-specific challenges: developing lightweight consensus protocols that preserve decentralization while accommodating vehicular mobility, designing privacy-preserving mechanisms that balance transparency with user anonymity, and creating efficient revocation schemes that maintain immutability's benefits while enabling dynamic credential management. By explicitly connecting open research questions to blockchain's core properties, this paper guides researchers toward solutions that leverage blockchain's strengths while mitigating its limitations in VANET contexts.

Ultimately, this paper challenges researchers to critically assess whether blockchain optimally solves specific VANET problems rather than adopting it merely for popularity. When blockchain is appropriate, its deployment should genuinely address concrete challenges through deliberate exploitation of its unique properties. This work aims to catalyze more thoughtful and impactful research, advancing both blockchain technology and vehicular networks.

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